In-depth Analysis for Genk

Leveraging Data Analytics for Efficient OR Planning at ZOL

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Introduction

Efficient management of operating rooms (ORs) requires an understanding of how planned surgical activity aligns with real-world execution. This document provides an in-depth, site-specific analysis for **Campus Genk**, serving as an analytical extension of the previous exploratory study. The goal is to systematically address the key research questions formulated in the project description:

Research questions

- **Planned versus observed timing**
  - How well do the planned operating durations correspond to the actual durations?
  - What is the typical deviation between the planned start time and the actual entry into the operating room?
  - Which specialisms or procedure types show the largest differences?
  - How do these deviations scale relative to the planned duration (proportional or percentage-wise)?
- **Adherence to working hours**
  - How often do procedures run beyond regular working hours (08:00 to 16:30)?
  - What proportion of OR activity occurs in the evening or at night?
  - How do these patterns differ between campuses?
- **Room allocation**
  - How often are procedures performed in a different room than originally planned?
  - Are these reallocations concentrated in specific rooms or specialisms?
- **Urgency and scheduling**
  - How are urgent and non-elective cases distributed across campuses and time periods?
  - To what extent do urgent surgeries overlap with elective activity and cause scheduling disruptions?

Each of the following sections addresses one of these themes in depth, focusing on the local characteristics of the Genk campus.

Unique values and data richness

The dataset captures a high degree of heterogeneity in surgical practice. A total of 60895 unique patient IDs are recorded, spread across 1295 distinct procedure codes and 1276 procedure names. Surgeon and anesthesiologist coverage is broad, with 195 and 207 unique identifiers respectively. These numbers reflect the diversity of clinical practice at ZOL and underscore the importance of developing grouping strategies to obtain actionable insights. Table 1 provides an overview of the number of unique values per variable.

Table 1: Number of unique values per variable.

Variable	UniqueValues
PatientID	60895
ProcedureCode	1295
ProcedureName	1276
AnesthesiologistID	207
HeadSurgeonID	195
LengthOfStay	166
PlannedOR	18
ActualOR	18
Weekday	7
Year	4
AdmissionType	2

Surgical procedure durations

Operating time is one of the most critical variables for planning. Procedure durations display wide variation, both within and across procedure types. Percentile analysis shows that 95% of all procedures finish within 253 minutes, 98% within 346 minutes, 99% within 411 minutes, and 99.9% within 630 minutes. While the majority of procedures are considerably shorter, the long right hand tail in the distribution indicates the presence of outliers and highly complex cases that drive variability in scheduling. Figure 1 illustrates the distribution of procedure durations, and Table 2 summarizes the percentile values.

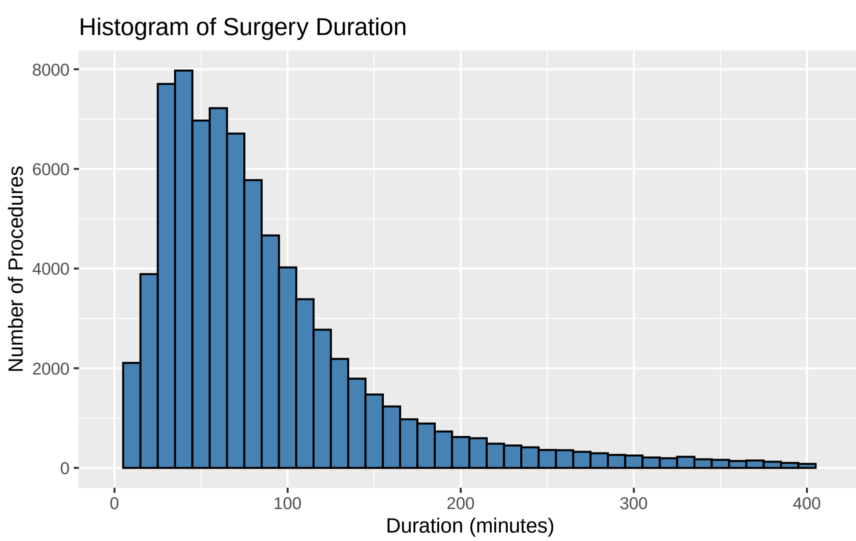

Figure 1: Histogram of Surgery Duration

Table 2: Percentiles of surgical procedure duration (in minutes).

Percentile	Duration (minutes)
95%	253
98%	346
99%	411
99.9%	630

Looking at the most frequent procedures illustrates the substantial variation in operating times across case types. Total hip replacement is among the longest and most resource intensive procedures, with an average duration of 81.6 minutes and a median of 76 minutes. Similarly, robot assisted total knee prosthesis procedures have an average duration of 114.6 minutes and a median of 112 minutes, reflecting their higher technical complexity. In contrast, several high volume procedures are considerably shorter. EXTRACTIE WHT has an average duration of 35.2 minutes and a median of 34 minutes, while EXTRACTIES average 36.7 minutes with a median of 34 minutes. Procedures such as URETERORENOSCOPIE and HNP-MICRO also show relatively shorter durations compared to the most complex cases, with median values of 36 and 91 minutes respectively. This combination of shorter standardized interventions and longer complex surgeries highlights the operational challenge of constructing efficient operating room schedules that must accommodate both high throughput activity and time intensive cases. Table 3 summarizes the ten most frequent procedure types.

Table 3: Top 10 most frequent procedure types with mean and median duration

ProcedureName	Count	Mean	Median
HEUP / TOTALE HEUPPROTHESE	3509	81.6	76
EXTRACTIE WHT	2998	35.2	34
HNP-MILD	1923	105.1	99
EXTRACTIES	1741	36.7	34
SECTIO	1732	70.9	70
TOTALE KNIE PROTHESE ROBOT ASSISTED ROSA	1331	114.6	112
LAP.GALBLAAS	1216	75.3	71
EXPLORATIEVE LAPARASCOPIE	1066	94.7	85
URETERORENOSCOPIE	973	41.7	36
HNP-MICRO	956	93.3	91

Admission type and urgency

The majority of procedures in the dataset are performed in a planned setting. Ambulatory surgery accounts for 42.3%, and inpatient hospitalization for 57.7%, while emergency admissions are not separately represented in the admission type data. The urgency classification shows a similar pattern: 85.4% of procedures are elective, and 14.6% are non elective. Although relatively small in number, the latter category remains important from a planning perspective, as urgent procedures may disrupt elective schedules and lead to cancellations or overtime. Table 4 and Table 5 summarize both distributions.

Table 4: Frequency table for Admission Type

AdmissionType	Count	Percentage
HOS	45798	57.7
DAG	33554	42.3

Table 5: Frequency table for Urgency Classification

UrgencyType	Count	Percentage
electief	67736	85.4
niet-electief	11616	14.6

Temporal distribution of activity

Surgical activity is spread consistently across weekdays. Thursdays account for 20.0 percent of weekly activity, Wednesdays for 19.9 percent, Fridays for 19.7 percent, Tuesdays for 18.6 percent, and Mondays for 17.9 percent. Weekend days show limited but non negligible activity, with 2.1 percent of procedures performed on Saturdays and 1.8 percent on Sundays. Annual trends reveal steady growth, with the number of procedures increasing from 22133 in 2022 to 23738 in 2024. At the time of data extraction, 9906 procedures in 2025 represented 12.5 percent of the total. Together, Table 6 and Table 7 summarize the distribution of surgical activity across weekdays and years.

Table 6: Frequency table for Day of the Week (sorted by descending count)

Weekday	Count	Percentage
Do	15844	20
Wo	15814	19.9
Vr	15615	19.7
Di	14750	18.6
Ma	14232	17.9
Za	1654	2.1
Zo	1443	1.8

Table 7: Frequency table for Year

Year	Count	Percentage
2022	22133	27.9
2023	23575	29.7
2024	23738	29.9
2025	9906	12.5

In summary, procedure durations show strong variability with a long tailed distribution driven by complex surgeries. The majority of procedures are elective, while non elective cases still have an important impact on schedule stability. Activity is distributed evenly across weekdays and has grown consistently in recent years. These findings form the baseline against which more advanced performance metrics such as start and end deviations, utilization rates, and predictive models can be developed in subsequent sections of this report.

How accurately does planning reflect reality in the OR?

Are planned and observed durations aligned?

The comparison between planned and observed surgery durations shows a broad spread around the one to one line. Many procedures lie close to the diagonal, indicating that their observed durations correspond reasonably well to the planned values. At the same time, a substantial number of points fall above or below this line, showing that both underestimation and overestimation occur frequently. Most deviations appear as longer than planned surgeries, visible in the dense area below the red dashed line. The dispersion increases for longer procedures, reflecting greater uncertainty for complex cases. Figure 2 displays this relationship.

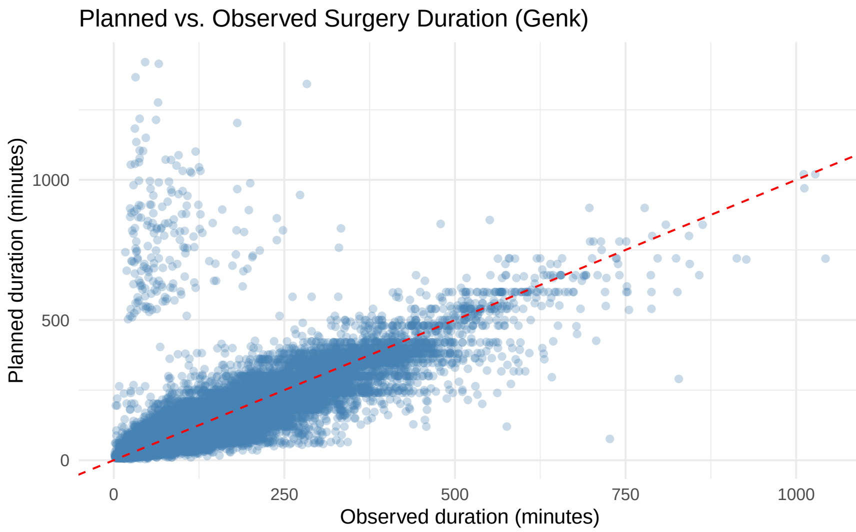

Figure 2: Planned vs. Observed Surgery Duration (Genk)

On average, surgeries at Genk show meaningful variation between planned and actual durations. Among procedures that last longer than planned, the mean deviation is 22.6 minutes with a median of 14 minutes, while shorter cases deviate on average by 21.2 minutes with a median of 12 minutes. The standard deviations of 27.5 minutes and 52.4 minutes highlight the presence of both moderate deviations and substantial outliers. While most differences remain within reasonable limits, a small number of extreme values extend the overall range.

Almost half of all surgeries (45.7%) take longer than planned, with an interquartile range of 23 minutes, while 54.3% finish earlier, with an interquartile range of 19 minutes. This distribution shows that deviations occur in both directions, but the spread is larger for cases that finish earlier. The considerable variability within each category indicates that although overall planning accuracy is reasonable at the aggregate level, individual durations remain difficult to predict. Improving the consistency of duration estimates, particularly for procedures that frequently deviate, could help reduce daily schedule variability and limit delays across the operating list. Table 8 provides a detailed overview of these differences.

Table 8: Difference between actual and planned duration (minutes), separated by direction

direction	Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9	Percent
Longer than planned	22.6	14	27.5	23	1	651	73	133	245.7	45.7%

direction	Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9	Percent
Shorter than planned	21.2	12	52.4	19	0	1374	58	122	856	54.3%

Expressing deviations as a percentage of the planned duration provides a standardized view across short and long procedures. Most cases fall within about ±40 percent of the planned time, indicating a generally acceptable level of scheduling accuracy in routine practice. The histogram peaks slightly below zero, showing that surgeries are on average completed a bit earlier than planned. The shape is roughly symmetrical, although the right tail extends further, indicating that large overruns occur less frequently but can be substantial when they appear. This pattern suggests that while most procedures are reasonably well estimated, a smaller group of cases contributes disproportionately to the overall variability in duration accuracy. Figure 3 shows the full distribution.

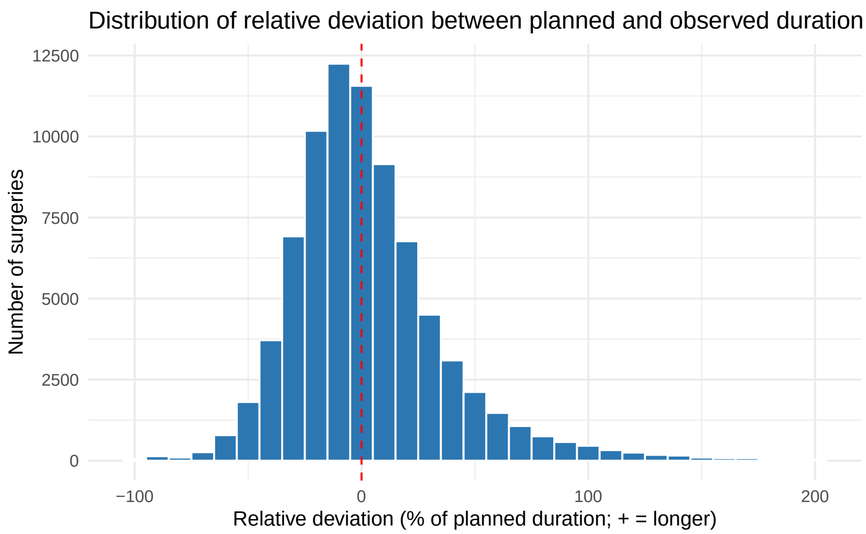

Figure 3: Distribution of relative deviation between planned and observed duration

Duration deviations differ markedly between admission types.

Inpatient, HOS, surgeries have an average deviation of 5.1 percent and a median of minus 1.3 percent, suggesting that most procedures are close to plan. The coefficient of variation of 7.8 indicates substantial variability between cases, implying that some procedures last much longer or shorter than expected.

Ambulatory, DAG, surgeries are the most predictable group. Their mean deviation is 2.8 percent, the median is minus 3.3 percent, and the coefficient of variation of 13.9 mainly reflects the fact that even moderate absolute differences translate into larger percentages for short procedures. These results show that standardized short-stay procedures are generally estimated accurately in the planning process. Table 9 summarizes these patterns.

Table 9: Relative duration deviation and coefficient of variation by admission type (Genk)

AdmissionType	n	Mean relative deviation (%)	Median relative deviation (%)	SD (%)	CV (%)
HOS	45796	5.1	-1.3	40.1	7.8
DAG	33553	2.8	-3.3	39.2	13.9

A similar pattern appears when comparing elective and non elective surgeries. Non elective procedures have an average relative deviation of 13 percent and a median of 3.1 percent, with a coefficient of variation of 4, highlighting their unpredictable character. Elective operations show smaller deviations, with an average of 2.6 percent and a median of minus 2.7 percent. Their coefficient of variation of 14 results from the influence of a small number of extreme outliers rather than consistent misestimation. Overall, these findings confirm that while elective cases generally follow expected durations well, non elective surgeries remain the main source of variability in the operating schedule. Table 10 summarizes these results.

Table 10: Relative duration deviation and coefficient of variation by urgency type (Genk)

UrgencyType	n	Mean relative deviation (%)	Median relative deviation (%)	SD (%)	CV (%)
niet-electief	11613	13	3.1	52.7	4
electief	67736	2.6	-2.7	36.9	14

The mean relative deviation between planned and observed durations remains broadly stable over the observed years. In 2022 the average deviation is 4.5 percent, decreasing slightly to 4.3 percent in 2023. A further slight decrease appears in 2024, reaching 4.2 percent, while the partial data for 2025 show a more pronounced decrease to 2.9 percent.

This pattern indicates that planning accuracy at Genk has remained relatively consistent over time, with only moderate year to year fluctuations. The differences are small and likely reflect normal operational variation rather than systematic changes in planning quality. Overall, the stability of the yearly averages suggests that the underlying scheduling process is well established, even though individual level variability continues to affect day to day accuracy. Figure 4 illustrates these trends.

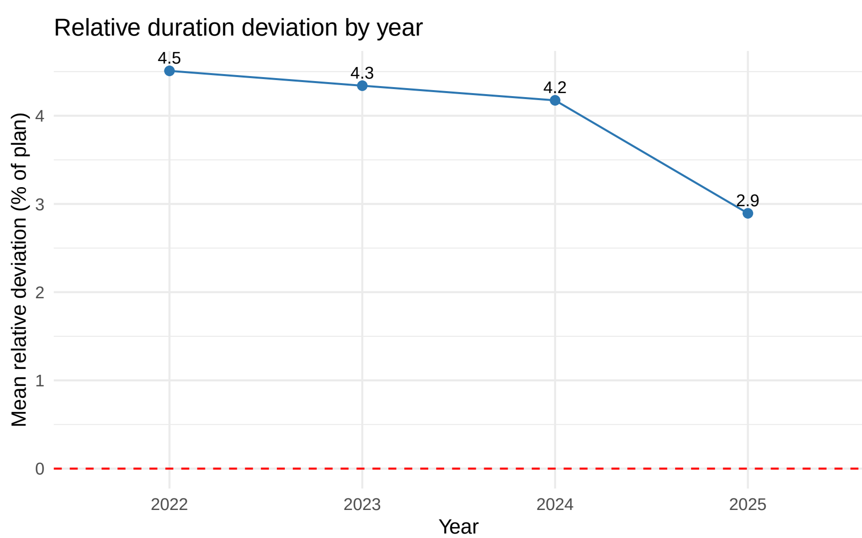

Figure 4: Relative duration deviation by year

The relative deviation between planned and observed durations remains stable throughout the regular workweek, with mean deviations ranging from 2.9 percent to 4.6 percent. This consistency indicates limited variation in planning accuracy across Monday to Friday. Small fluctuations, such as the slightly higher averages on Wednesday, likely reflect differences in case mix rather than structural scheduling issues.

A clear contrast appears during the weekend. On Saturdays and Sundays the mean deviations rise sharply to 23.1 percent and 23.7 percent, far above weekday levels. These higher discrepancies are driven by the smaller and more unpredictable volume of urgent cases. Weekend surgeries are typically non elective, often requiring flexible resource allocation and showing less predictable durations. The weekday weekend contrast therefore highlights the stability of elective scheduling during standard operating days and the inherent variability associated with emergency work at the end of the week. Figure 5 summarizes these differences.

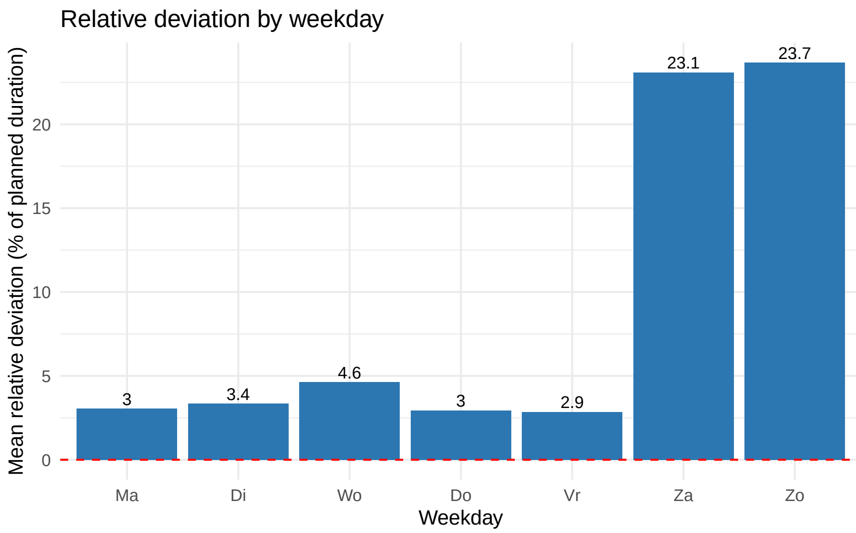

Figure 5: Relative deviation by weekday

Variation in deviation differs considerably across operating rooms. The largest average deviations are concentrated in the complex surgery rooms of the GO block, most notably GO10, where the mean absolute deviation is 54.5 minutes and 50.3 percent of cases run longer than planned. Other high variability rooms such as GO09, GO12, and GO13 also show average deviations between 28.3 and 34.5 minutes, reflecting the higher uncertainty typical of lengthy or technically demanding procedures.

By contrast, rooms such as GO14, GO01, and GO16, mainly associated with shorter and more standardized interventions, display mean deviations below 17 minutes and a relatively balanced share of shorter and longer cases. These differences illustrate how the degree of procedural complexity and specialization strongly influences planning precision. Table 11 provides the detailed overview.

Table 11: Duration deviations per operating room, ordered by mean absolute deviation

ActualOR	n	Mean abs dev	Longer (%)	Mean relative dev (longer)	Shorter (%)	Mean relative dev (shorter)
GO10	1742	54.5	50.3	26.1	48.6	-18.5
GO09	2637	34.5	48	31.2	50.6	-20
GO08	1777	29.3	51.2	40.2	46.7	-21.1
GO12	2884	29.1	48.6	25.4	49.5	-17.7
GO13	3298	28.3	48.4	28.8	49.8	-17.7
GO05	4886	26.6	49.2	35.3	48.7	-20.2
GO04	4518	23.8	44.6	32.1	53	-20.2
GO11	7481	23.8	56.8	45.4	41.5	-22
GO02	3502	21.9	47.3	31.4	50.8	-18.8
GO03	4093	21.9	44.3	28.4	53.4	-18.4
GO06	5217	20.3	38.9	31.5	58.2	-23
GO15	4323	19.8	41.4	27	56.3	-21
GO17	5480	18.2	37.6	31.8	60	-23.4

ActualOR	n	Mean abs dev	Longer (%)	Mean relative dev (longer)	Shorter (%)	Mean relative dev (shorter)
GO07	4658	17.9	45.7	29	52	-19.4
GO18	5293	17.6	39.5	27.7	58.2	-22
GO16	4098	17.5	42.9	22.3	55.4	-18.1
GO01	6577	16.2	48.3	35.5	48.4	-23
GO14	6885	13	44.4	37.5	51.6	-22.7

Among the most extreme deviations in surgery duration, longer and shorter cases occur in almost equal proportions. Around 50.4 percent of the top one percent of deviations correspond to procedures that last longer than planned, while 49.6 percent finish earlier than expected. This contrasts with earlier observations based on a different dataset, where overruns dominated more clearly.

Although both directions are represented among the extreme cases, their operational impact remains different. Early finishes generally create limited idle time, whereas overruns can delay subsequent cases and increase the likelihood of overtime. The near balance between longer and shorter cases in the current extreme deviations suggests that large underestimations and overestimations both contribute meaningfully to schedule variability, reinforcing the need to monitor procedures prone to substantial deviations in either direction. Table 12 summarizes these results.

Table 12: Direction of deviation among the 1% most extreme cases (Genk)

Direction	n	Percentage
Longer than planned	405	50.4%
Shorter than planned	398	49.6%

A closer look on the coefficient of variation

The monthly coefficient of variation for planning deviation shows clear fluctuations over time without a consistent long term upward trend. As shown in Figure 6, early 2022 begins at relatively low variability levels near 1.1 to 1.3. Through the remainder of 2022 the values move mostly between 1.3 and 1.9, indicating modest dispersion in planning deviations.

During 2023 the coefficient rises intermittently, with several months reaching values between about 2.0 and 2.7, although lower values around 1.3 also occur. This results in a more irregular pattern rather than a continuous increase. In early 2024 the coefficient peaks briefly at around 2.8, marking the largest monthly value in the series. After this spike the metric decreases again and fluctuates between roughly 1.6 and 2.5 during the remainder of 2024.

In 2025 the values stabilise around 2.1 to 2.4, showing moderate variability. Overall, the pattern alternates between higher and lower dispersion rather than showing a systematic upward trend. The temporary peak around 2.8 suggests that certain periods experience substantially more unpredictable durations, while most months remain within a moderate range of variability.

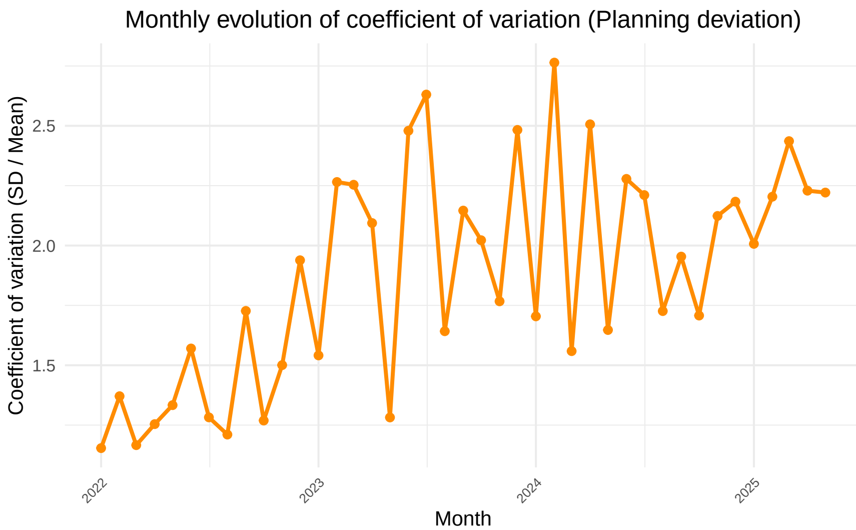

Figure 6: Monthly evolution of coefficient of variation (Planning deviation)

Variability by planned duration group

Variation in surgical timing can be examined from two complementary perspectives that describe different aspects of performance. The first is the coefficient of variation of observed duration, calculated as the standard deviation divided by the mean of the observed surgical times. This indicator expresses how dispersed the recorded durations are relative to their average. In a dataset such as this one, which contains a large number of both short and long operations, a high coefficient of variation mainly reflects the broad spread in case duration across procedure types rather than instability within individual surgeries.

The second perspective is the coefficient of variation of the planning deviation, which measures the consistency of planning accuracy. It is computed as the standard deviation of the absolute deviation between planned and observed time, divided by the mean of that deviation. This metric focuses on how variable the alignment between planned and realised durations is, regardless of how long the surgeries themselves last.

Comparing both perspectives helps disentangle two components of variability: the structural heterogeneity in procedure duration and the variability in how precisely these durations are planned. Understanding how these two aspects evolve provides insight into whether deviations primarily originate from the diversity of surgical activity or from variation in scheduling accuracy. The following table summarises these indicators across five duration categories and serves as a baseline for the more detailed, category-specific analyses presented in the subsequent subsections. All values are reported in Table 13, which provides the exact CV measures for both observed durations and planning deviations.

The summary statistics confirm a clear scaling effect in relative variability. The coefficient of variation of observed durations decreases as planned duration increases: procedures planned for less than 30 minutes have a CV of 0.5420733, procedures in the 31–60 minute group show 0.4345528, the 61–90 minute group 0.3468322, the 91–180 minute group 0.3500428, and the longest category over 180 minutes shows 0.4177433. This pattern is consistent with the fact that shorter cases amplify small absolute fluctuations when expressed relative to their mean, while longer cases distribute absolute variation over a broader time base.

For the planning deviation, coefficients of variation are considerably higher across all groups, reflecting the larger proportional spread in the differences between planned and actual duration. These values range from 1.1904840 in the shortest category to 1.1016421 in the 31–60 minute group, 1.0800735 in the 61–90 minute group, 0.9095045 in the 91–180 minute group, and 1.8358983 in procedures lasting more than 180 minutes. This pattern suggests that longer and more complex procedures, while not necessarily less stable in execution, exhibit greater variation in planning accuracy.

Table 13: Summary of mean duration, SD, and CV by planned duration group (Observed vs. Planning deviation)

Group	n	Mean duration	SD duration	CV duration	Mean planning	SD planning	CV planning
<30	8749	27.51766	14.91659	0.5420733	8.747628	10.41391	1.1904840
31–60	23387	47.93860	20.83185	0.4345528	12.639672	13.92439	1.1016421
61–90	19371	78.28971	27.15339	0.3468322	18.097620	19.54676	1.0800735
91–180	20533	121.38255	42.48909	0.3500428	26.067988	23.70895	0.9095045
> 180	7309	273.35956	114.19413	0.4177433	64.916541	119.18017	1.8358983

Together, these figures show that relative variability in observed duration is largely structural and reflects case mix composition, whereas variability in planning deviation captures the irregularity in scheduling precision. The distinction between the two metrics provides the foundation for the next analyses, which explore how these patterns manifest within each duration category and across operating rooms.

The monthly evolution of the coefficient of variation for observed durations shows stable and clearly separated patterns between the planned duration groups. Short procedures in the <30 minute category consistently display the highest relative variability, with values generally between 0.5 and 0.7. Procedures in the 31–60, 61–90, and 91–180 minute groups cluster at lower levels, mostly between 0.3 and 0.5, while operations longer than 180 minutes follow a similar but slightly more dispersed pattern. These differences remain consistent over the entire observation period, indicating that the proportional dispersion of observed surgical duration within each category is structurally stable. Figure 7 illustrates these monthly patterns across duration groups.

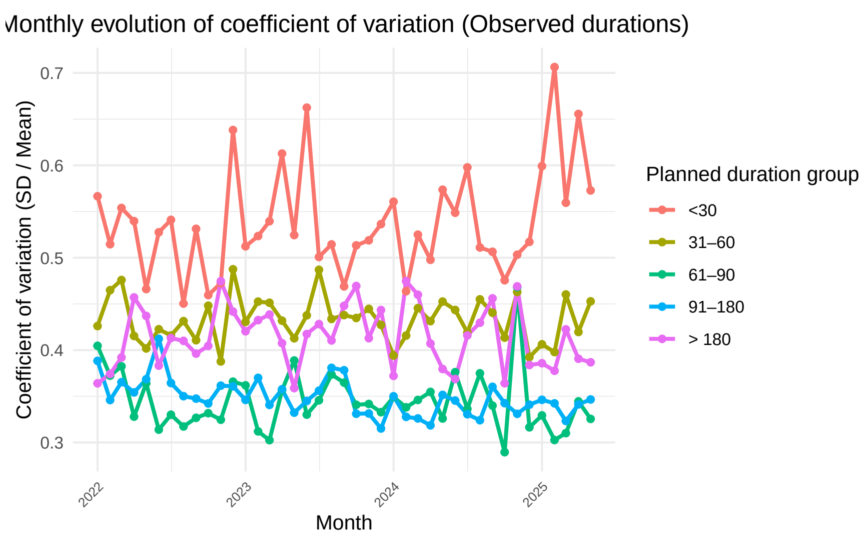

Figure 7: Monthly evolution of coefficient of variation (Observed durations)

The coefficient of variation of planning deviation shows more pronounced fluctuations over time than the coefficient of variation of observed durations. The shortest procedures (<30 minutes) and the longest procedures (>180 minutes) exhibit the largest variability, with several peaks reaching values between 1.4 and 2.3. These higher fluctuations likely reflect temporary shifts in case mix or periods with a larger share of procedures that are inherently more difficult to predict. The mid-range duration groups (31–90 minutes) display lower and more stable coefficients of variation, indicating that planning accuracy for these procedures is more consistent, generally around 0.9 to 1.2. Figure 8 shows how these patterns evolve monthly across the duration groups.

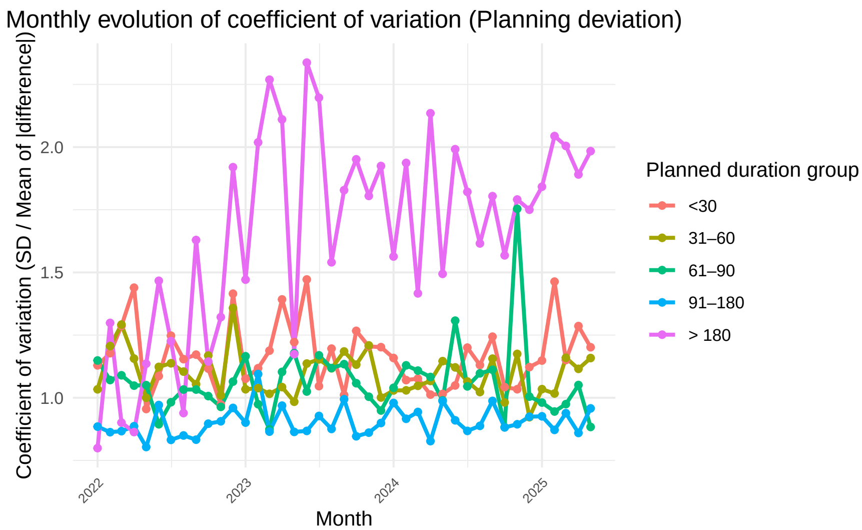

Figure 8: Monthly evolution of coefficient of variation (Planning deviation)

Viewed together, the two figures show that the relative variability of observed surgery durations remains structurally stable across months and duration categories, whereas the variability in planning deviation fluctuates more strongly over time, especially for the shortest and longest procedures. These contrasting patterns indicate that execution itself is consistent, while planning accuracy is more sensitive to shifts in case mix or procedure complexity. The subsequent sections examine these patterns in greater detail by analysing each duration group separately and comparing the distribution of coefficients of variation across operating rooms.

Surgeries planned under 30 minutes

Short procedures, typically representing high volume and standardized interventions, show moderate relative variability across operating rooms. For this category, coefficients of variation of observed durations range roughly between 0.32 and 0.65, which reflects the proportional impact of small absolute timing fluctuations on short cases. These values are expected for brief procedures, where even limited variation translates into relatively large coefficients once expressed relative to their mean duration. Table 14 summarises these indicators for all rooms with at least eleven surgeries in this duration group.

The distribution across rooms suggests a broadly homogeneous pattern. Units such as GO17, GO14, and GO13 show higher CV values between 0.56 and 0.65, while others including GO01, GO06, and GO04 show lower variability, with values between 0.32 and 0.46. These differences are small in absolute terms and mainly reflect differences in the mix of short procedures performed in each room rather than substantive disparities in timing consistency.

For the coefficient of variation of planning deviation, values are systematically higher and span a wider range, with most rooms falling between approximately 0.79 and 1.37. This confirms that relative fluctuations in planning accuracy exceed those associated with the observed durations themselves. The lowest CV deviation values appear in rooms such as GO03 and GO12, while higher values are found in rooms including GO02 and GO13.

Overall, short procedures are characterized by consistent execution times and broadly comparable planning accuracy across operating rooms. The relatively high coefficients primarily reflect proportional scaling effects inherent to short interventions rather than substantial operational variability. Subsequent sections examine whether similar patterns hold for procedures of longer planned duration, where absolute variability becomes more dominant and planning precision potentially more challenging.

Table 14: Coefficient of variation per operating room for surgeries planned <30 minutes (only rooms with >10 surgeries shown)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
GO14	2380	19.68	12.02	0.6108	6.524	7.679	1.177
GO01	2006	32.43	11.25	0.3468	8.77	8.992	1.025
GO17	938	20.31	13.28	0.654	6.935	8.494	1.225
GO18	698	26.28	11.53	0.4388	7.099	7.631	1.075
GO11	641	38.72	19.78	0.5108	15.91	17.36	1.091
GO06	495	29.95	11.5	0.3839	8.374	8.541	1.02
GO15	306	25.09	14.33	0.5711	8.683	8.842	1.018
GO05	226	32.46	13.73	0.4231	10.19	10.25	1.006
GO07	224	32.79	11.91	0.3631	9.54	9.638	1.01
GO04	179	32.15	14.88	0.4627	10.15	12.95	1.277
GO16	160	27.88	14.87	0.5332	9.35	10.77	1.151
GO02	126	35.8	17.05	0.4761	11.94	16.05	1.345
GO03	97	41.11	13.17	0.3202	15.28	12.07	0.7902
GO08	91	40.8	18.93	0.4639	17.04	16.34	0.9586
GO13	63	38.97	22.04	0.5655	14.46	19.82	1.371
GO09	44	41.3	22.98	0.5564	18.14	20.3	1.119
GO10	38	35.37	15.43	0.4363	11.71	12.09	1.033
GO12	37	37.81	13.71	0.3625	13.76	12.26	0.8908

Surgeries planned 31–60 minutes

Procedures with a planned duration of 31–60 minutes show moderate relative variability across operating rooms. For this category, coefficients of variation of observed durations cluster around 0.31–0.49 for most rooms, which reflects the proportional spread seen in Table 15. The highest values appear in GO02 (0.4864), GO13 (0.4791), and GO12 (0.4442), while several rooms, including GO09 (0.3122), GO01 (0.4398), and GO06 (0.4247), show lower variability. These differences likely reflect variation in the mix of procedures performed rather than structural differences in workflow.

For the coefficient of variation of planning deviation, values are again higher across all rooms, ranging roughly between 0.97 and 1.38. As shown in Table 15, the lowest CV_deviation values appear in rooms such as GO09 (0.97) and GO15 (0.975), while larger or multipurpose rooms such as GO13 (1.381) and GO05 (1.216) show higher values. This pattern indicates that relative variation in planning accuracy exceeds the variation in the durations themselves, but the dispersion remains modest and relatively uniform.

Taken together, these results show that procedures planned for 31–60 minutes display consistent execution times and moderate variation in planning accuracy across operating rooms. The observed coefficients largely follow proportional scaling effects typical of medium-length procedures rather than substantial operational differences.

Table 15: Coefficient of variation per operating room for surgeries planned 31–60 minutes (only rooms with >10 surgeries shown)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
GO01	3083	40.74	17.92	0.4398	10.77	11.99	1.114
GO06	2851	42.04	17.85	0.4247	12.29	12.11	0.985
GO11	2627	56.4	24.31	0.4311	16.44	18.28	1.111
GO14	2433	44.78	16.39	0.3661	10.62	11.26	1.061
GO07	1666	47.4	19.49	0.4111	11.95	11.99	1.003

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
GO18	1473	44.87	18.42	0.4105	11.42	11.22	0.9826
GO05	1373	56.27	24.62	0.4376	14.69	17.86	1.216
GO04	1367	53.93	22.38	0.415	13.54	15.15	1.12
GO15	1106	46.75	17.84	0.3815	11.13	10.85	0.975
GO02	1045	48.26	23.47	0.4864	13.92	15.97	1.147
GO17	988	44.73	19.5	0.436	12.22	12.06	0.9871
GO03	782	53.15	20.22	0.3804	12.85	14.62	1.138
GO16	733	46.99	18.32	0.3899	12.06	12.11	1.004
GO08	639	50.06	18.39	0.3674	12.17	12.3	1.011
GO09	610	56.64	17.68	0.3122	13.2	12.81	0.97
GO13	271	57.58	27.59	0.4791	16.1	22.23	1.381
GO12	187	52.41	23.28	0.4442	13.74	17	1.238
GO10	153	53.95	21.56	0.3996	15.37	15.21	0.9901

Surgeries planned 61–90 minutes

Procedures planned between 61 and 90 minutes represent a balanced segment of the surgical workload, combining moderately complex inpatient cases with standardized elective interventions. Within this group, the coefficient of variation of observed durations is relatively low and stable across operating rooms. As shown in Table 16, most rooms display CV values between 0.26 and 0.42, indicating proportionally consistent execution times once procedures of this length begin. The limited dispersion suggests that mid-length operations, often supported by standardized preparation and anaesthesia routines, show predictable timing.

Differences between operating rooms remain modest. The lowest observed CVs occur in GO12 (0.2561), GO17 (0.2651), and GO18 (0.2856), while higher values appear in GO05 (0.3962) and GO10 (0.4381). The highest variability within this category is found in GO08 (0.5879), reflecting the broader mix of procedures handled in that room. Overall, these differences seem driven mainly by case mix rather than operational inconsistency.

The coefficient of variation of planning deviation follows a similar but slightly broader distribution. Most rooms show CV_deviation levels between 0.80 and 1.19, indicating moderate variation in planning accuracy. As shown in Table 16, rooms such as GO16 (0.8741) and GO12 (0.8713) show lower CV values, while others, including GO05 (1.189) and GO13 (1.075) display higher dispersion in planning deviation. These patterns suggest that although mean estimates are generally accurate, deviations vary more widely for some case mixes.

Overall, surgeries in the 61–90 minute category show stable and proportional duration patterns with limited variation across operating rooms. Planning accuracy shows moderate dispersion but remains broadly consistent across the complex. Compared with shorter procedures, these mid-length cases appear less affected by proportional scaling effects and provide a relatively steady basis for daily scheduling.

Table 16: Coefficient of variation per operating room for surgeries planned 61–90 minutes (only rooms with >10 surgeries shown)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
GO11	2050	84.44	33.39	0.3954	23.58	24.52	1.04
GO17	1962	76.98	20.41	0.2651	15.05	13.78	0.9162
GO18	1856	74.89	21.39	0.2856	15.59	13.74	0.8809
GO07	1786	75.23	21.91	0.2912	14.98	14.86	0.9923
GO05	1520	79.92	31.66	0.3962	21.07	25.05	1.189
GO04	1515	74.98	26.85	0.3581	17.54	19.58	1.116
GO03	1400	75.72	22.17	0.2928	15.11	14.84	0.9818

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
GO14	1307	72.77	23.36	0.3211	16.73	15.44	0.9228
GO15	1145	75.62	24.31	0.3215	16.38	15.84	0.9673
GO02	788	80.36	26.54	0.3302	18.1	19.01	1.05
GO16	753	77.32	23.4	0.3026	16.62	14.53	0.8741
GO12	749	89.79	22.99	0.2561	18.3	15.94	0.8713
GO13	713	91.77	31.19	0.3399	22.72	24.43	1.075
GO06	615	71.05	23.73	0.3341	18.47	14.9	0.8066
GO01	471	73.77	30.83	0.418	20.22	22.17	1.096
GO09	330	81.04	33.93	0.4187	22.2	26.44	1.191
GO08	321	90.02	52.92	0.5879	28.53	45.74	1.603
GO10	90	88.62	38.83	0.4381	28.76	29.24	1.017

Surgeries planned 91–180 minutes

Surgeries with a planned duration between 91 and 180 minutes include a broad mix of major inpatient operations and complex elective cases. Within this category, the coefficient of variation of observed durations is moderate, ranging mostly between 0.23 and 0.45 across operating rooms, as shown in Table 17. The lowest CV values are found in GO16 (0.2346) and GO12 (0.3086), while slightly higher values appear in GO04, GO09, and GO11, where CV_duration reaches around 0.37 to 0.39. These differences remain limited and suggest proportionally consistent variability in execution time despite the broader absolute duration range.

The coefficient of variation of planning deviation shows similarly narrow dispersion. Most rooms fall between 0.75 and 1.02, with the lowest values seen in GO17 (0.6998) and GO18 (0.7718). The highest CV_diff values occur in GO10 (1.024) and GO08 (1.005), but these remain close to the overall pattern. This consistency indicates stable planning accuracy for longer procedures, even when case mix becomes more heterogeneous.

Across operating rooms, both indicators show restrained variability, implying that once surgeries reach this duration range, timing performance becomes comparatively stable. In contrast to shorter procedures, proportional variation in duration and planning deviation becomes smaller relative to total time. These mid- to long-duration operations therefore represent one of the most consistent segments of the surgical program in terms of timing reliability.

Table 17: Coefficient of variation per operating room for surgeries planned 91–180 minutes (only rooms with >10 surgeries shown)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
GO16	2395	112.9	26.47	0.2346	18.14	16.27	0.8969
GO11	1946	116.4	45.39	0.39	29.9	28.24	0.9443
GO15	1627	123.6	38.74	0.3135	25.42	22.34	0.8789
GO13	1617	124	40.48	0.3265	23.88	21.32	0.8927
GO03	1571	120	34.64	0.2888	24.07	21.49	0.8926
GO17	1497	100.5	40.75	0.4056	28.93	20.24	0.6998
GO12	1213	121.8	37.6	0.3086	22.92	20.62	0.8998
GO02	1170	130.9	37.81	0.2889	24.03	21.82	0.908
GO18	1163	110.4	38.98	0.3531	26.85	20.73	0.7718
GO05	1115	132.5	48.81	0.3684	28.91	28.66	0.9915
GO09	951	143.7	54.47	0.3791	32.6	31.89	0.9781
GO07	823	122.6	39.54	0.3224	23.95	19.38	0.809
GO04	802	125.5	46.94	0.374	30.15	25.05	0.8308
GO01	761	118.7	43.56	0.367	29.83	22.56	0.7564
GO06	660	148.6	45.41	0.3056	26.74	21.85	0.8171

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
GO14	660	110.1	39.96	0.363	25.55	22.45	0.8789
GO08	437	144.4	56.77	0.3931	37.08	37.26	1.005
GO10	125	127.4	57.67	0.4526	38.82	39.76	1.024

Surgeries planned over 180 minutes

Procedures with a planned duration exceeding three hours represent the most complex and resource intensive part of the surgical program. Within this group, the coefficient of variation of observed durations remains moderate, ranging from 0.30 to 0.65 across operating rooms, as shown in Table 18. Most rooms fall between 0.35 and 0.45, indicating broadly consistent proportional variability even for long and heterogeneous procedures. The lowest CV values appear in GO10 (0.2963), GO16 (0.3113), and GO12 (0.3467), while higher values are observed in GO04, GO05, and GO02. GO01 shows the highest CV_duration at 0.6498, although interpretation is limited due to its smaller case count.

The coefficient of variation of planning deviation is generally close to or above one. Most rooms fall between 1.27 and 2.08, with the lowest values in GO10 (1.273) and GO12 (1.375). Higher values appear in several rooms, with GO14 (2.514) and GO06 (2.334) among the highest. These differences reflect the wide absolute spread in deviations inherent to long procedures, rather than systematic planning inconsistencies.

Overall, surgeries longer than three hours display stable and proportionally uniform duration patterns across the operating complex. Observed durations show limited relative variability, and planning deviations exhibit consistent proportional spread. The higher coefficients observed in a few rooms likely result from concentration of rare or high complexity procedures rather than persistent scheduling issues.

Table 18: Coefficient of variation per operating room for surgeries planned > 180 minutes (only rooms with >10 surgeries shown)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
GO10	1336	346.3	102.6	0.2963	63.37	80.65	1.273
GO09	702	254.8	94.69	0.3717	62.33	102.2	1.64
GO12	698	278	96.37	0.3467	56.19	77.26	1.375
GO04	655	259.8	116.6	0.4486	55.76	93.2	1.671
GO05	652	237	107.3	0.4528	66.1	131.4	1.988
GO13	634	272.6	100.2	0.3677	52.65	91.53	1.739
GO06	596	257.5	94.61	0.3675	63.09	147.2	2.334
GO02	373	307.1	132.8	0.4323	48.7	66.97	1.375
GO08	289	244.3	84.54	0.346	60.33	82.96	1.375
GO01	256	296.7	192.8	0.6498	91.95	196.5	2.137
GO03	243	229	91.97	0.4017	79.14	161	2.034
GO11	217	239	108.3	0.4531	84.74	136.1	1.606
GO07	159	236	116	0.4915	93.25	179.3	1.922
GO15	139	193	74.56	0.3863	76.14	148.6	1.952
GO14	105	236	93.55	0.3963	87.01	218.7	2.514
GO18	103	208.7	77.56	0.3716	111.1	231.4	2.082
GO17	95	210.6	90.17	0.4282	85.74	166.5	1.942
GO16	57	214.3	66.7	0.3113	93.72	207.7	2.216

Relationship between variability and mean operative duration

Examining how variability scales with operative duration provides insight into whether dispersion in surgical times increases proportionally with the length of a procedure. By analysing this relationship at the level of procedure types, it becomes possible to distinguish structural scaling effects from differences in planning performance. The following figures present three complementary perspectives: the standard deviation of operative duration, the coefficient of variation of observed duration, and the coefficient of variation of planning deviation. Together, they describe how both absolute and relative variability evolve as surgeries become longer or more complex.

The first figure shows a strong positive linear relationship between mean operative time and its standard deviation, as illustrated in Figure 9. As the average duration increases, the absolute spread of recorded times expands almost proportionally. This scaling effect is expected, since longer procedures allow for greater absolute variation while maintaining a similar proportional spread. For instance, procedures with a mean duration near one hundred minutes typically show standard deviations around forty to fifty minutes, while operations exceeding three hundred minutes often vary by more than one hundred minutes. This pattern confirms that absolute variability grows with case length without implying greater relative unpredictability.

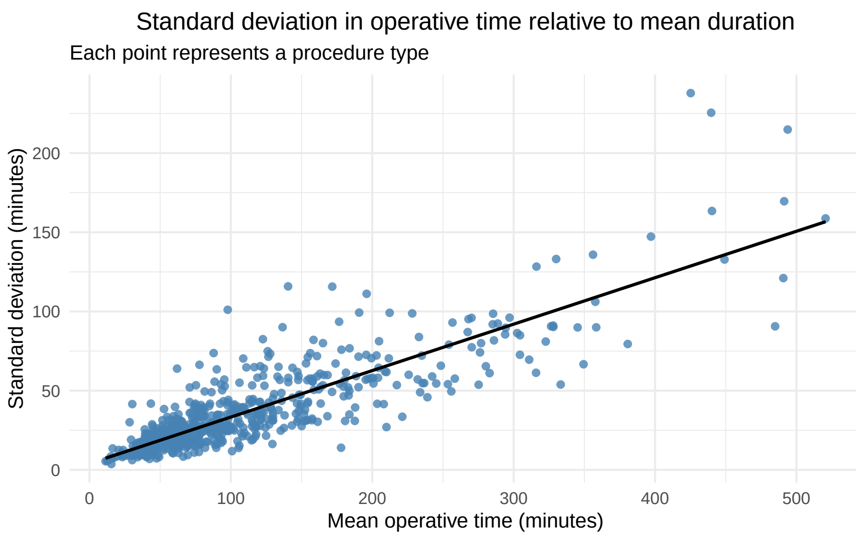

Figure 9: Standard deviation in operative time relative to mean duration

When variability is expressed relative to mean duration, a clear inverse pattern emerges, as illustrated in Figure 10. The coefficient of variation decreases gradually as procedures become longer, indicating that proportional dispersion diminishes with case length. Short procedures often show CV values above 0.6, while most long operations fall below 0.4. This relationship reflects a simple scaling property, where longer procedures display greater absolute variability but proportionally more stability compared with their expected duration.

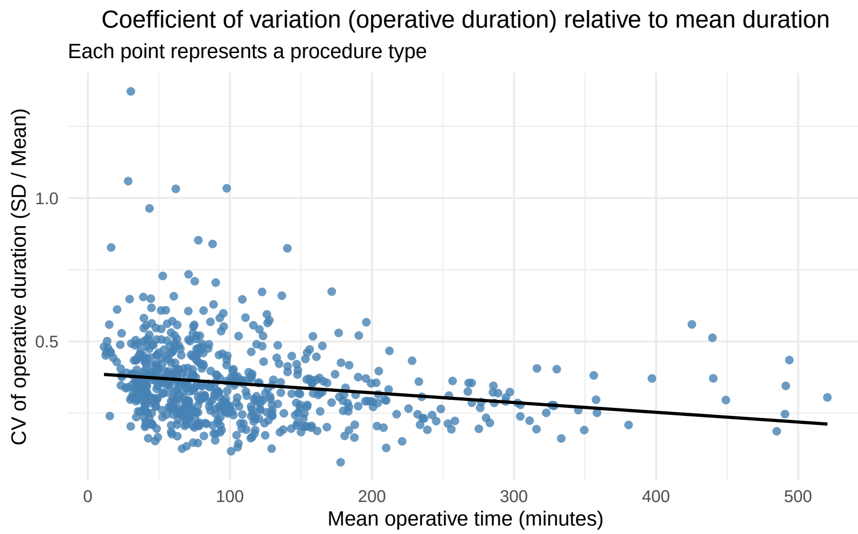

Figure 10: Coefficient of variation (operative duration) relative to mean duration

A similar downward trend is visible when plotting the coefficient of variation of planning deviation against mean duration, as shown in Figure 11. For shorter procedures, the relative spread in planning deviations is wide, with CV values sometimes exceeding 3.0 to 5.0. These high coefficients occur because small absolute deviations represent large proportional differences when the planned duration is short. As procedure length increases, this proportional effect diminishes, and the CV of planning deviation stabilizes around 1.0 or lower. The negative slope therefore reflects the same scaling principle observed for operative duration, rather than a systematic improvement in planning accuracy.

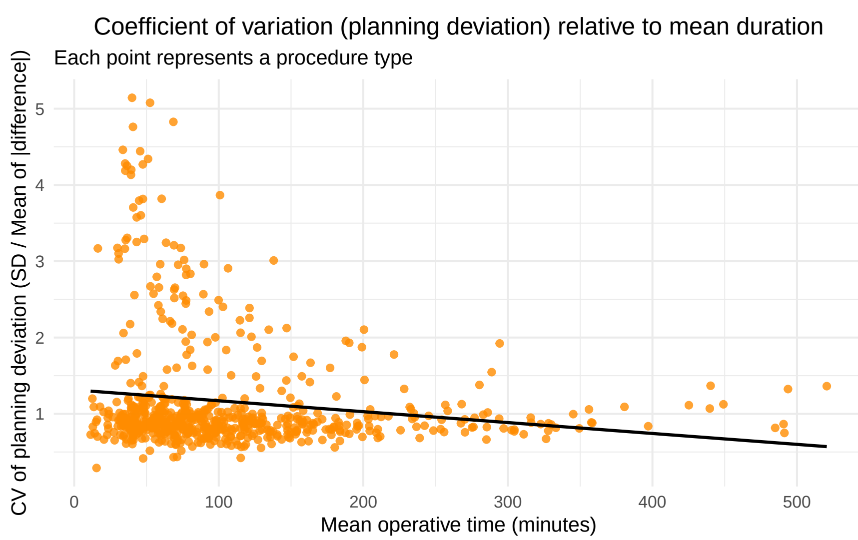

Figure 11: Coefficient of variation (planning deviation) relative to mean duration

Taken together, the three figures show that while absolute variability in surgical time increases with case length, relative variability tends to decrease. This pattern indicates that proportional uncertainty is greatest for short procedures, where small timing differences represent a large share of the total duration, and least for long surgeries, where absolute deviations are diluted across extended operating periods.

How large are start-time delays, and when do they happen?

Start-time deviations show a wide spread, with late starts occurring more often and with greater magnitude than early ones, as summarized in Table 19. About 67.4% of all surgeries begin later than scheduled, with an average delay of 74.6 minutes and a median of 28 minutes. Early starts occur in 32.6% of cases, averaging 31.9 minutes ahead of plan and a median of 10 minutes. The dispersion is considerable, as indicated by standard deviations of 163.4 minutes for late starts and 68.9 minutes for early ones. A few extreme outliers reach delays above 1,800 minutes, emphasizing substantial day-to-day variability.

These results confirm that punctuality at the beginning of the surgical schedule remains a key operational challenge, since late starts tend to cascade through the remainder of the day and reduce overall room efficiency.

Table 19: Difference between actual and planned start time (absolute minutes), separated by direction

direction	Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9	Percent
Late start	74.6	28	163.4	51	1	1863	419	899	1317	67.4%
Early start	31.9	10	68.9	27	0	1366	122	368	720	32.6%

The distribution of start-time differences is centered around zero, as shown in Figure 12, with a clear peak at small deviations, indicating that most surgeries start close to their planned time. However, the shape of the histogram is slightly skewed toward positive values, showing that late starts are more frequent than early ones. Most cases begin within approximately half an hour of schedule, but the gradual widening of the distribution toward the right reveals that a smaller subset experiences significant delays.

The presence of a long right-hand tail confirms that occasionally, extreme delays still occur. These outliers have a limited frequency but can affect overall operating room throughput and schedule stability. The symmetrical appearance around the center, combined with a mild skew, suggests that deviations are not random but follow a consistent pattern of minor delay accumulation across the day.

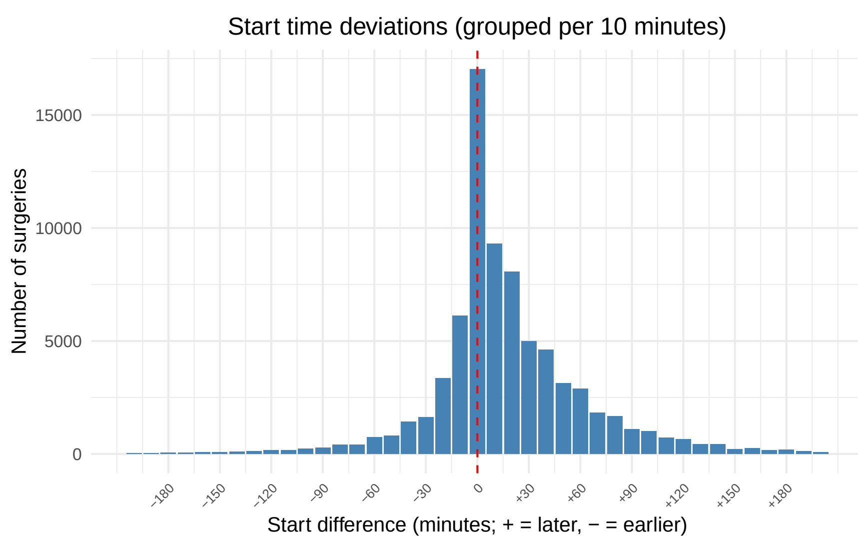

Figure 12: Start time deviations (grouped per 10 minutes)

Average start-time delays show moderate variation across weekdays, as illustrated in Figure 13, ranging from 37.3 to 42.6 minutes during regular working days. Monday has a mean delay of 42.6 minutes, Tuesday 39.9

minutes, while Wednesday reaches 40.3 minutes. Thursday and Friday follow with 37.3 and 38.4 minutes, respectively. These small differences indicate that start-time punctuality is relatively consistent throughout the week, with only minor fluctuations that are unlikely to reflect structural scheduling issues.

In contrast, weekend operations display noticeably longer average delays, reaching 46.4 minutes on Saturday and 44.5 minutes on Sunday. These larger values likely relate to the lower number of scheduled cases and the more variable timing of urgent or ad hoc procedures. Overall, the pattern indicates that weekday starts are fairly well aligned with their planned times, whereas weekend activity tends to be less predictable.

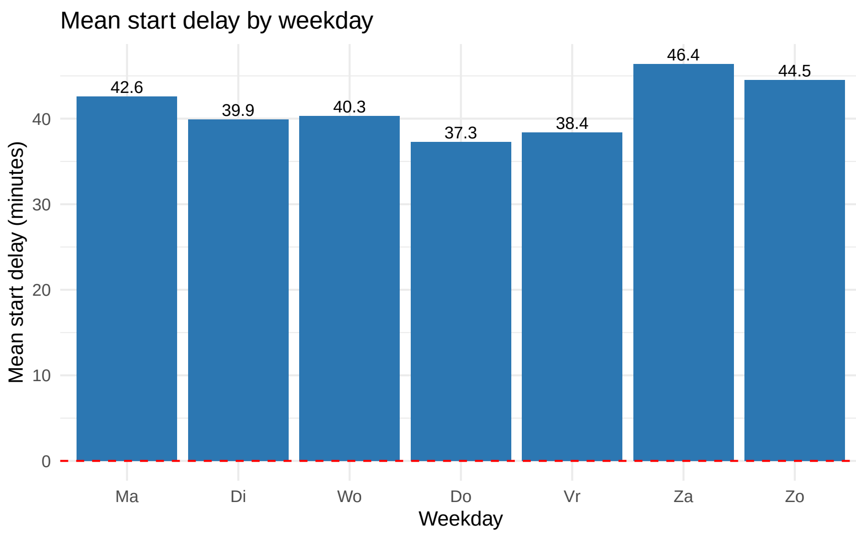

Figure 13: Mean start delay by weekday

Late starts occur more frequently than early starts on every day of the week, as summarized in Table 20. Across regular weekdays, about two thirds of surgeries begin later than planned, with mean delays between 70.9 and 79.3 minutes and median delays around 26 to 28 minutes. Early starts are less common, representing roughly 28.7 to 31.2% of all cases, and typically begin between 29.7 and 39.4 minutes before the scheduled time. These small variations between weekdays indicate that overall punctuality during the workweek is relatively stable, with no clear day standing out as an outlier.

During the weekend, start times become less predictable. On both Saturday and Sunday, about 77.5 to 77.6% of surgeries start late, and early starts drop to about 21.3% of cases. Mean delays increase to 75.5 minutes on Saturday and 73.2 minutes on Sunday, with median delays of 44 and 41 minutes, respectively. This pattern suggests that weekend operations are subject to greater variability, likely reflecting the smaller number of scheduled procedures and the higher share of urgent cases.

Table 20: Start-time deviation patterns per weekday (minutes).

Weekday	n	Late starts (%)	Mean delay	Median delay	Early starts (%)	Mean lead	Median lead
Ma	14232	66.6	79.3	28	30.9	-33.1	-12
Di	14750	66.4	76.1	26	31.2	-34.1	-12
Wo	15814	68.8	70.9	27	28.7	-29.7	-11
Do	15844	66.2	71.8	27	31.2	-32.8	-12
Vr	15615	66.8	75.5	27	30.6	-39.4	-13
Za	1654	77.6	75.5	44	21.3	-57	-23
Zo	1443	77.5	73.2	41	21.3	-57.3	-21

Start-time reliability varies widely between operating rooms, as shown in Table 21. The proportion of late starts ranges from 46.1% to 82.4%, indicating that punctuality is unevenly distributed across the OR complex. Some rooms, such as GO11 and GO14, stand out with very high shares of late starts, 82.4% and 78.7%, respectively. In GO11, the mean delay among late cases reaches 319.1 minutes, driven by a small number of extreme outliers that strongly influence the average.

Other rooms, including GO14, GOP1, and GO02, also show predominantly late starts (around 70–79%) but with more moderate mean delays of 37 to 42 minutes. At the opposite end, rooms such as GO10 display a much more balanced distribution, where early starts occur roughly as often as late ones. These contrasts suggest that certain rooms operate under more stable and predictable timing conditions, while others experience recurring or structural delays. Overall, start-time deviations appear concentrated in a limited set of operating rooms, where occasional long postponements account for much of the overall variability in punctuality.

Table 21: Start-time deviation patterns per operating room (minutes).

ActualOR	n	Late starts (%)	Mean delay	Median delay	Early starts (%)	Mean lead	Median lead
GO11	7482	82.4	319.1	103.5	16.2	-34	-12
GO14	6885	78.7	37.3	24	19.5	-35.2	-15
GO01	6577	75.1	38	25	23	-34.4	-15
GO08	1777	71.9	44.9	30	25.8	-47.2	-10
GO02	3502	70.3	42.2	25	26.5	-33.2	-12.5
GO07	4658	70	43.5	23	27.3	-34.3	-12
GO05	4886	68.3	54.3	31	29.4	-46.6	-14
GO03	4094	65.9	44.6	26	30.7	-37.3	-13
GO09	2637	64.5	48.9	22	32.3	-34.7	-9
GO04	4518	63.5	49.4	25	33.6	-35.5	-14
GO15	4323	63.1	40.5	26	34.5	-28.1	-11
GO16	4098	62.6	35	25	35.1	-22.5	-9
GO18	5293	62.3	36.9	24	35.9	-30.2	-12
GO17	5480	61.8	38.4	22	35.9	-36.4	-14
GO06	5217	61.7	42.1	24	35.2	-41.7	-17
GO12	2884	54.5	48.3	24	42	-32.3	-10
GO13	3298	53.7	48.6	26	42.5	-31.9	-11
GO10	1743	46.1	63.2	29	50.8	-31.4	-8

Which procedures deviate most from plan?

Procedures differ greatly in how accurately their duration is predicted, as shown in Table 22. The table lists the twenty procedure types with the largest absolute deviations between planned and observed time. Average deviations range from about 52 to more than 74 minutes, showing that for some procedure types, the planned duration can diverge considerably from actual performance.

The highest deviations are seen in procedures such as DEBULKING MET HIPEC, AVR, and DEBULKING, with mean absolute differences above 70 minutes. These gaps indicate that complex or less frequent interventions are particularly difficult to plan accurately. Procedures with moderate case counts, including COR.AORTA BYPASS GRAFT V. and P.L.I.F. MET BRAINLAB AERO, also show average deviations around 55 to 60 minutes, confirming that variability remains high even for procedures with more frequent occurrence.

Across the listed procedures, longer-than-planned cases are not consistently more common than shorter ones, suggesting no clear systematic tendency toward underestimation. Overall, the results highlight that planning accuracy strongly depends on procedure type, with predictable durations for routine operations but substantial uncertainty for specialized or infrequently scheduled cases.

Table 22: Top 20 procedures by average absolute duration deviation (Genk).

ProcedureName	n	Mean abs dev (min)	Longer (%)	Mean + dev (min)	Shorter (%)	Mean - dev (min)
DEBULKING MET HIPEC	76	74.4	44.7	77.2	55.3	-72.2
AVR	118	73.7	44.1	67.5	55.9	-78.6
DEBULKING	75	72.2	33.3	55.2	64	-84
FEMUR FRAKTUUR LFN	41	68.5	56.1	73.1	43.9	-62.5
TROMBECTOMIE	60	66.6	63.3	75.9	35	-52.8
FEMORO TIBIALE BYPASS	54	63.3	63	67.7	35.2	-58.8
CABG OFF PUMP	127	60.4	29.1	42.3	70.9	-67.9
THORACOSCOPIE BULLECTOMIE	30	59.8	33.3	50.9	63.3	-67.7
ABDOMINAAL ANEURYSMA	80	59.5	50	60.6	48.8	-59.8
COR.AORTA BYPASS GRAFT V.	654	59.5	53.1	58.4	45.7	-62.4
P.L.I.F. MET BRAINLAB AERO	229	54.8	53.7	61.4	44.1	-49.5
HALSEVIDEMENT + MONDTUMOR	32	54.2	31.2	33.7	68.8	-63.5
FEMORO POPLITEALE BYPASS	177	53.6	49.2	61.6	50.8	-45.8
AORTA BIFEMORALE GREFFE	37	53.4	56.8	52.9	43.2	-54.1
PORT ACCES MITRALISKLEP	124	53.4	44.4	54.9	53.2	-54.7
AV FISTEL VERWIJDEREN	33	53.2	75.8	34.4	24.2	-112
LAP. GEASS. VAGINALE HYSTRECTOMIE MET LYMPHE ADENOPATHIE	38	52.5	21.1	34.6	78.9	-57.2
GESTEELDE FLAP	37	52.4	35.1	57.4	64.9	-49.7
FRACTUUR MANDIBULA	67	52.2	62.7	46.7	35.8	-64
RAL ANTERIOR	113	52.1	45.1	53.7	54.9	-50.8

The scatter plot in Figure 14 shows an inverse relationship between procedure volume and average deviation. Procedures that occur more frequently tend to have lower mean absolute deviations, while rare procedures show much greater variability in duration. This pattern is visible in the downward trend of the points as procedure volume increases.

High-volume procedures, which appear toward the right of the chart, cluster at lower deviation values, indicating more stable and predictable planning performance. In contrast, low-volume procedures on the left display a much wider range of deviations that can exceed 60 minutes. The relationship suggests that familiarity and repetition contribute to more accurate duration estimates, whereas infrequent procedures are harder to predict due to limited historical data. Overall, the figure highlights that case volume is an important factor in explaining the variability of planning accuracy across procedure types.

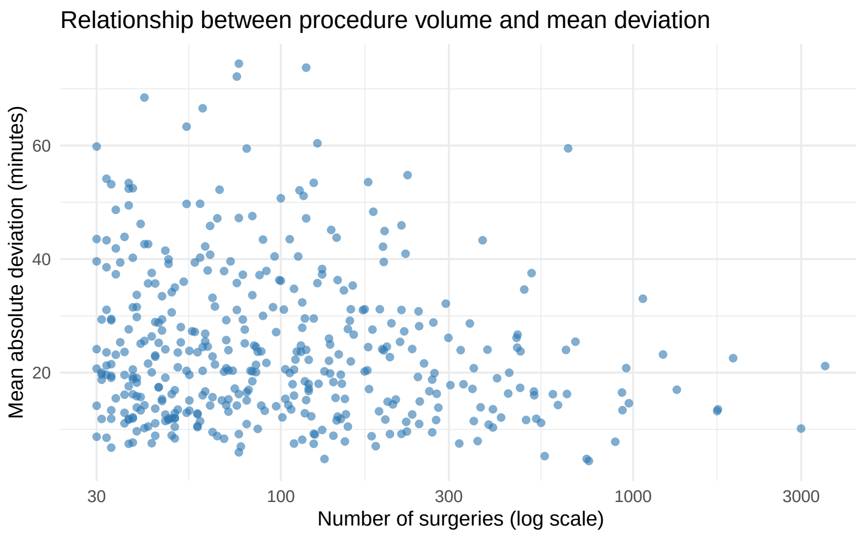

Figure 14: Relationship between procedure volume and mean deviation

The distribution of duration deviations across surgeons reveals notable variation in planning accuracy, as shown in Table 23. Among those with at least fifty recorded procedures, the average absolute deviation ranges from 30.4 to 68.3 minutes, while the average relative deviation spans 17.2% to 53.2%. A few surgeons show comparatively high mean deviations above 55 minutes, whereas most fall between 30 and 40 minutes, indicating that planning performance differs across individuals but remains within a moderate range for the majority.

No clear relationship appears between the number of recorded cases and the size of the deviation, suggesting that factors beyond case volume influence the observed variation. Overall, the results show that duration deviations are not uniformly distributed across surgeons, emphasizing that planning accuracy can vary meaningfully even within a shared operational environment.

Table 23: Top 15 surgeons by average absolute and relative duration deviation (pseudonymized).

Surgeon	n	Mean absolute dev (min)	Mean relative dev (%)
Surgeon 1	375	68.3	23.8
Surgeon 2	251	59.6	22.3
Surgeon 3	418	57.1	19.1
Surgeon 4	481	55.3	20.5
Surgeon 5	111	51.0	17.2
Surgeon 6	189	37.9	53.2
Surgeon 7	580	36.6	30.9
Surgeon 8	165	35.7	29.3
Surgeon 9	683	34.9	25.3
Surgeon 10	1801	33.3	24.4

Surgeon	n	Mean absolute dev (min)	Mean relative dev (%)
Surgeon 11	663	32.3	26.4
Surgeon 12	50	32.0	19.0
Surgeon 13	1057	31.4	27.9
Surgeon 14	422	30.5	23.7
Surgeon 15	264	30.4	28.7

Do surgeries stay within regular working hours?

How often do cases extend beyond their assigned time window?

Around 9.7 percent of all surgeries at Genk continue beyond the scheduled shift end of the operating team, as shown in Table 24. On average, these extended cases run about 60.3 minutes beyond the planned shift, with a median of 39 minutes. Most overruns are therefore limited in length, but a smaller subset of cases last considerably longer, as reflected by the 95th percentile of 197 minutes. These outliers typically represent complex or unpredictable procedures that continue well into the next operational block.

The results indicate that overtime is not confined to the day shift but is a recurring feature across the full 24-hour cycle. While the majority of surgeries conclude within their scheduled working periods, the 9.7 percent share of extended cases points to regular spillover between adjacent shifts. The substantial variation in overtime duration also suggests that although most overruns are moderate, a few extended sessions contribute disproportionately to the overall evening and night workload.

Table 24: Overall share and duration of after-hours surgeries (Genk).

Total	Surgeries with overtime	Percentage	Mean Overtime	Median Overtime	P95 Overtime
79352	7729	9.7	60.3	39	197

The distribution of overtime duration, shown in Figure 15, is strongly right-skewed. Most extended cases end within the first 60 minutes beyond the scheduled shift, and the highest frequency occurs between 0 and 40 minutes of overtime. Beyond that, the number of cases declines steadily, with only a small fraction exceeding 200 minutes. A few extreme cases surpass 300 minutes, but these are rare.

This pattern confirms that while overtime is a recurring operational feature, it is typically limited in duration. The long right tail illustrates that a small group of prolonged procedures accounts for a disproportionate share of total overtime minutes. These outliers can heavily affect staff workload and room turnover late in the day, but they do not represent the typical case. Overall, overtime at Genk mainly consists of brief extensions of planned activity rather than frequent, sustained night work.

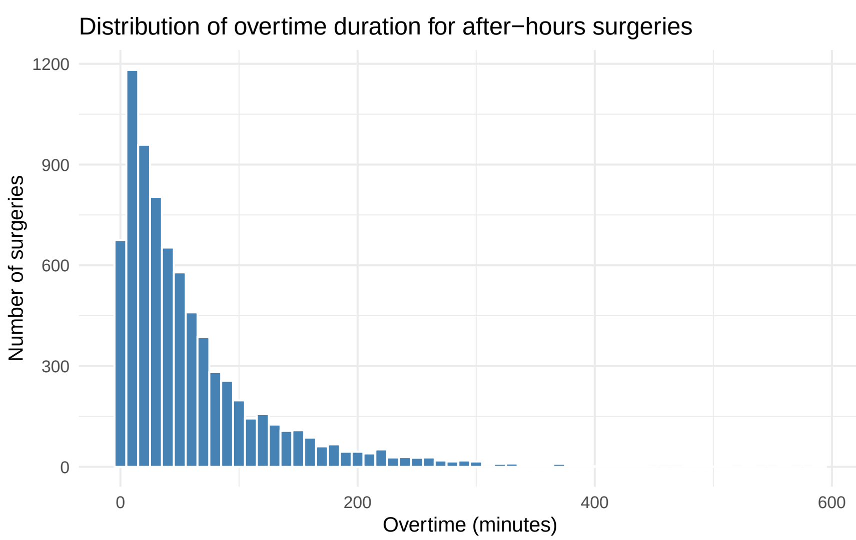

Figure 15: Distribution of overtime duration for after-hours surgeries

The share of surgeries extending beyond scheduled shift hours remains steady throughout the regular workweek, as shown in Figure 16, fluctuating between 8.8 and 9.9 percent from Monday to Friday. This indicates that weekday operations are largely contained within their planned timeframes, with only a small proportion of cases requiring additional time beyond shift boundaries.

A marked increase occurs during the weekend. On Saturday, 16.8 percent of surgeries continue beyond the scheduled shift end, and on Sunday, the proportion remains high at 15.5 percent. Although the overall surgical volume is lower during weekends, these values suggest that a greater share of procedures performed on those days exceed the available shift window. This pattern likely reflects the predominance of urgent or semi-urgent cases during weekends, which are less predictable in both timing and duration.

Overall, the weekday–weekend contrast highlights how overtime at Genk is structurally embedded in the hospital's continuous surgical service. While overtime during weekdays is relatively infrequent and well controlled, weekend operations more often extend into subsequent shifts due to the emergency-driven case mix.

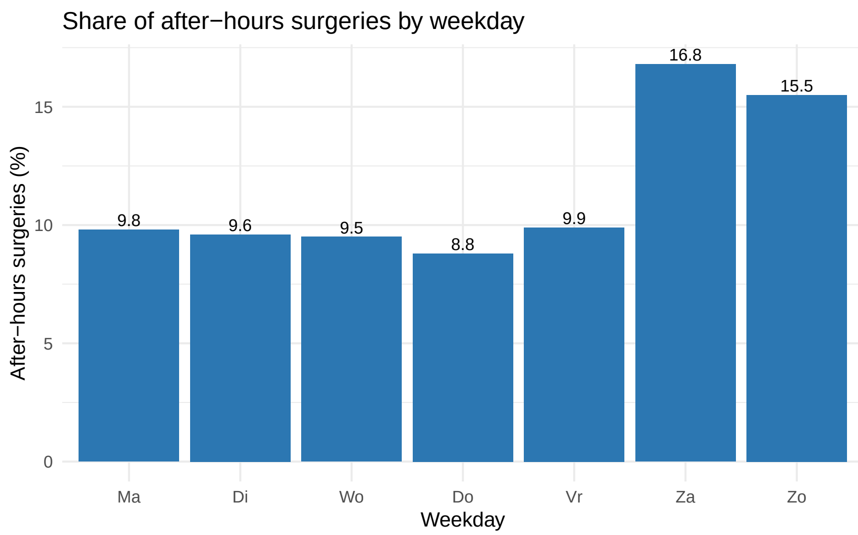

Figure 16: Share of after-hours surgeries by weekday

The proportion of surgeries extending beyond scheduled shift hours has shown a gradual decline over time, as illustrated in Figure 17. In 2022, about 10 percent of all surgeries at Genk exceeded their scheduled shift, compared with 10 percent in 2023, 9.7 percent in 2024, and 8.6 percent in the partial dataset for 2025.

These values represent the percentage of all procedures performed in each year that continued past the official end of their assigned shift. The downward trend therefore indicates a steady reduction in the relative frequency of overtime cases over the observed period.

Although the year-to-year changes are modest, the pattern suggests gradual improvement in adherence to scheduled shift boundaries, possibly reflecting smoother daytime throughput, more efficient list planning, or better alignment between case mix and available operating time. Overall, overtime remains a consistent but slowly diminishing part of operating room activity at Genk.

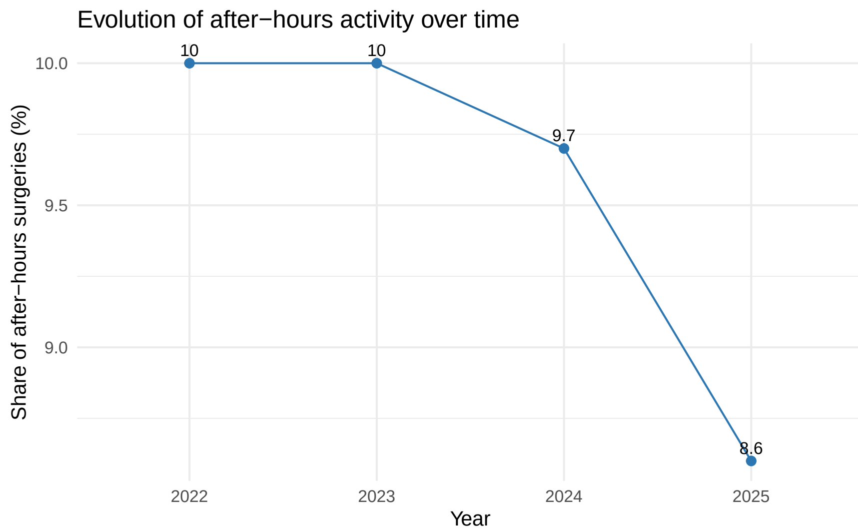

Figure 17: Evolution of after-hours activity over time

The proportion of surgeries extending beyond scheduled shift hours differs markedly across operating rooms, as summarized in Table 25. Most rooms show overtime rates between 5.8 and 13.7 percent, while a few stand out with substantially higher shares. GO10 records the highest proportion of overtime cases at 32.9 percent, with a mean duration of 154.2 minutes and a 95th percentile of 327.7 minutes, indicating frequent and occasionally prolonged extensions. Such high figures likely reflect the type of procedures performed in this room, which are often complex and less predictable in length.

Several other rooms, including GO09, GO12, and GO13, show overtime in roughly 13 to 16 percent of cases, with mean durations between 59.4 and 71.6 minutes. These values suggest that while most surgeries in these rooms finish within their allocated shift, a notable share still extends well into the next operational block.

In contrast, a larger group of rooms such as GO01, GO07, GO15, GO16, and GO17 show more moderate overtime frequencies below 10 percent, with average extensions typically between 36.9 and 57.8 minutes. The lowest rates occur in the smaller or more standardized rooms (GO14), where overtime is rare and mean durations remain short.

Overall, the data indicate that overtime at Genk is concentrated in a limited number of high-volume or complex operating rooms. While many rooms complete their scheduled activity within the planned shift most of the time, a considerable subset still experiences overtime in 10 to 15 percent of surgeries, showing that after-shift extensions remain a recurring part of daily operations rather than isolated exceptions.

Table 25: Operating rooms by after-hours activity (Genk).

ActualOR	Cases	Percentage of overtime surgeries	Mean Overtime	Median Overtime	P95
GO10	1743	32.9	154.2	142.5	327.7
GO09	2637	16.3	71.6	53	204.1
GO13	3298	13.7	61.3	39	173.6
GO08	1777	13.6	68.5	43	230.8
GO12	2884	13.3	59.4	41	176.4
GO05	4886	12.3	54.8	39	160.9
GO11	7482	11.7	55.1	40	156.2
GO02	3502	11.4	52.1	38	149.6
GO04	4518	10.7	57.4	40	155.8

ActualOR	Cases	Percentage of overtime surgeries	Mean Overtime	Median Overtime	P95
GO03	4094	9.6	45	33	142.8
GO06	5217	9.4	58.6	41	174.8
GO16	4098	8.9	36.9	26	101.8
GO15	4323	8.7	46.6	34	130.2
GO07	4658	6.9	38	28	103.8
GO17	5480	6.8	44.2	27	138.5
GO18	5293	6.5	41.6	28	127.5
GO01	6577	5.8	57.8	36	195.2
GO14	6885	3.5	31.4	19.5	87

When does overtime occur, and how heavy is it?

The distribution of overtime surgeries, shown in Figure 18, displays a clear concentration of cases ending shortly after 16:30. This peak primarily represents procedures that started during the regular day shift and finished later than planned. The number of surgeries declines rapidly as the evening progresses, indicating that most cases extending beyond their scheduled shift end within the first one to two hours afterward.

After 22:00, only a smaller number of surgeries continue to appear, suggesting that operations ending during the late evening or night are relatively infrequent. Very few procedures extend past midnight, and only isolated cases finish after 07:00.

Overall, the pattern demonstrates that overtime activity is dominated by operations that conclude soon after the end of the regular working day, while surgeries finishing later in the night represent a much smaller portion of total activity.

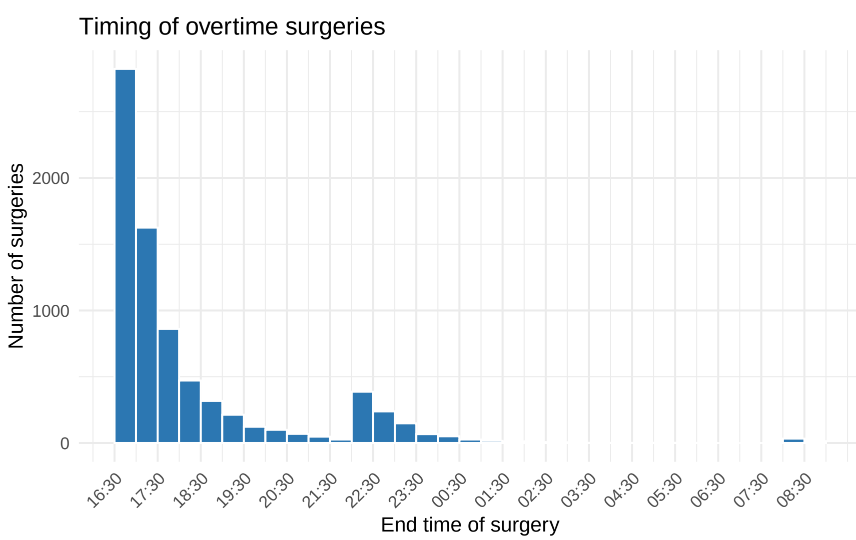

Figure 18: Timing of overtime surgeries

Average overtime durations are highly consistent across weekdays, as illustrated in Figure 19, with means ranging from 58.5 to 62.4 minutes from Monday to Friday. The median per weekday is clearly lower, between 36 and 41 minutes, which indicates a right-skewed distribution where a small number of long cases raises the mean.

On the weekend the pattern is similar. Saturday shows a somewhat higher mean (63.4 minutes) and a higher median (45 minutes). Sunday has a mean of 59.1 minutes and a median of 46 minutes. These values should not be interpreted as "shorter" or "longer" relative to regular daytime hours, since the weekend does not follow a standard weekday program. They simply reflect the overtime distribution of cases that meet the same per-shift definition.

Overall, once surgeries run past the shift boundary, the typical overrun is around 36 to 46 minutes (medians), while the average sits near one hour because of a small number of lengthy extensions.

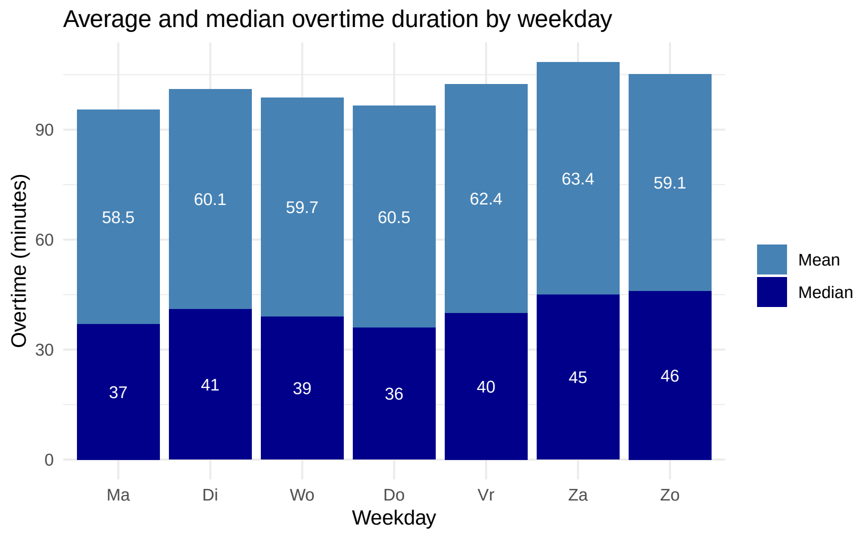

Figure 19: Average and median overtime duration by weekday

Average and median overtime durations per surgery show a stable pattern over the observed years, as illustrated in Figure 20. The mean varies only slightly, from 61.8 minutes in 2022 to 58.2 minutes in 2025, while the median remains consistently lower, between 37 and 40 minutes.

This consistent gap between the mean and median indicates that overtime durations are right-skewed, with a limited number of long cases pulling the mean upward. In most years, the typical overtime duration (median) is therefore around 37 to 40 minutes, whereas the average is closer to one hour.

The small changes over time suggest that the overall duration of overtime has remained stable. However, the average alone does not capture how variable these overruns are, two years can have the same mean but very different distributions. The persistent difference between the mean and median values highlights this variability and shows that a minority of longer cases continues to account for a disproportionate share of total overtime minutes.

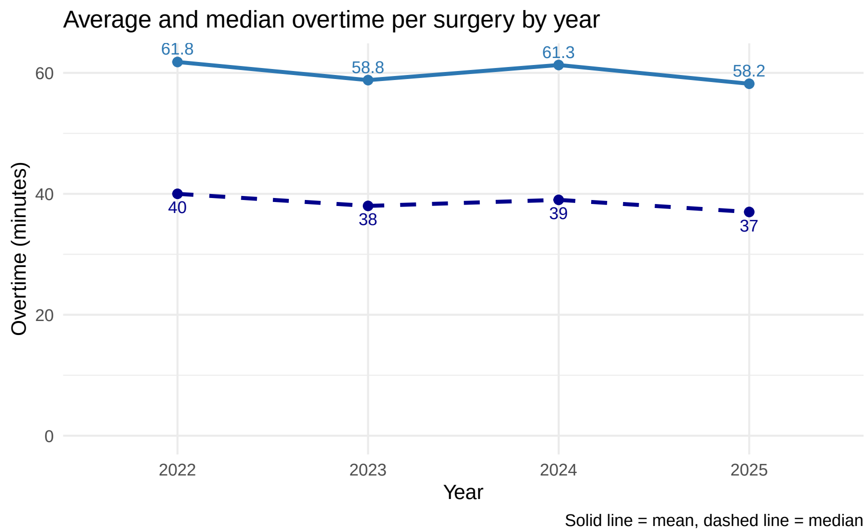

Figure 20: Average and median overtime per surgery by year

The distribution of overtime differs noticeably between shift types, as shown in Figure 21. The day shift (08:00–16:30) displays the widest spread, with most cases extending only slightly beyond the scheduled end but a small number continuing for several hours. This pattern explains why the day shift accounts for most overtime minutes overall.

In the evening shift (16:30–22:00), the spread is narrower and the majority of cases cluster close to zero, indicating that only a limited number of procedures continue far beyond the planned shift end.

The night shift (22:00–08:00) shows a similar pattern, with most observations concentrated around short extensions and only a few long outliers. The lower density compared with earlier shifts reflects the smaller total number of surgeries performed during this period.

Overall, the distributions illustrate that overtime is most variable during the day, when activity levels are highest, and more contained during the later shifts. The pronounced right tails in each group confirm that a limited number of long-running cases account for a substantial share of total overtime minutes.

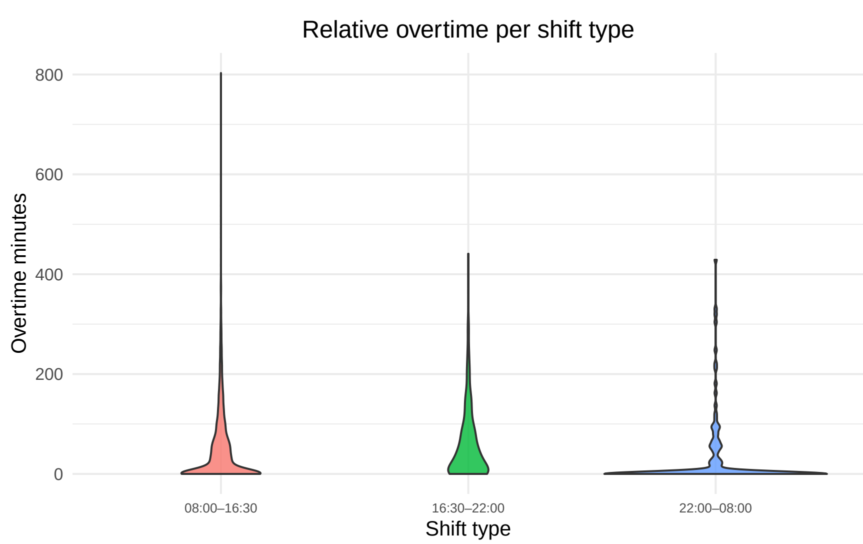

Figure 21: Relative overtime per shift type

How does Genk compare with Cathlab, Maaseik, and Lanaken?

The comparison between campuses shows clear differences in the extent of afterhours activity, as summarised in Table 26. At Genk, 9.7 percent of all surgeries extend beyond 16:30, compared with 5.8 percent at Cathlab, 2.8 percent at Maaseik, and only 0.5 percent at Lanaken. This gradient reflects both the higher surgical volume and the broader case mix at Genk and, to a lesser extent, at Cathlab, where more complex or lengthy procedures are scheduled.

Average overtime at Genk is also longest, with a mean of 60.3 minutes and a ninety fifth percentile of 197 minutes. Cathlab and Maaseik occupy an intermediate position, with mean overtime around 42 minutes and ninety fifth percentiles of 108.8 and 129 minutes respectively, while Lanaken shows clearly shorter extensions, with a mean of 15.7 minutes and a ninety fifth percentile of 48.5 minutes. These differences indicate that although evening work occurs on all campuses, it is most pronounced at the main site in Genk and, to a lesser degree, at Cathlab. Together, the results emphasize that Genk in particular carries the largest share of extended surgical operations within the ZOL network.

Table 26: Comparison of after-hours activity between campuses.

Campus	Total	AfterHour surgeries	Share of overtime surgeries	Average Overtime	Median Overtime	P95 Overtime
Genk	79352	7729	9.7	60.3	39	197
Cathlab	9215	536	5.8	41.9	33	108.8
Maaseik	53888	1492	2.8	42.1	29	129
Lanaken	69362	371	0.5	15.7	10	48.5

Are surgeries performed in their assigned operating rooms?

How frequently are surgeries relocated to a different room?

Room reallocations are relatively rare at Genk, as shown in Table 27. Out of 79352 recorded surgeries, around 0.7 percent were performed in a different operating room than originally planned. This low percentage indicates that room assignments are followed very consistently, with only a small number of last-minute adjustments required during daily operations.

Although infrequent, these relocations are operationally important because they typically involve reactive scheduling decisions and resource coordination. Even minor disruptions can have downstream effects on room preparation, staff planning, and turnover time. Tracking their frequency provides useful insight into the stability of the planning process and helps identify whether reassignments occur sporadically or tend to cluster in specific rooms or days.

Table 27: Overall frequency of surgeries relocated to a different room (Genk).

Total	Swapped (absolute)	Swapped (%)
79352	519	0.7

The share of relocated surgeries remains very low across all weekdays, as shown in Figure 22, generally fluctuating between 0.4 and 0.6 percent. This stability indicates that the room assignment process is robust during regular working days, with only minimal need for unplanned reallocations. The lowest rate is observed on Thursday (0.4%), while Tuesday, Wednesday, and Friday show slightly higher values, though still around 0.6 percent.

A clear increase appears during the weekend, where the share rises to 4.6 percent on Saturday and 3.5 percent on Sunday. These higher values likely reflect the smaller number of available rooms combined with a less predictable schedule during weekend operations.

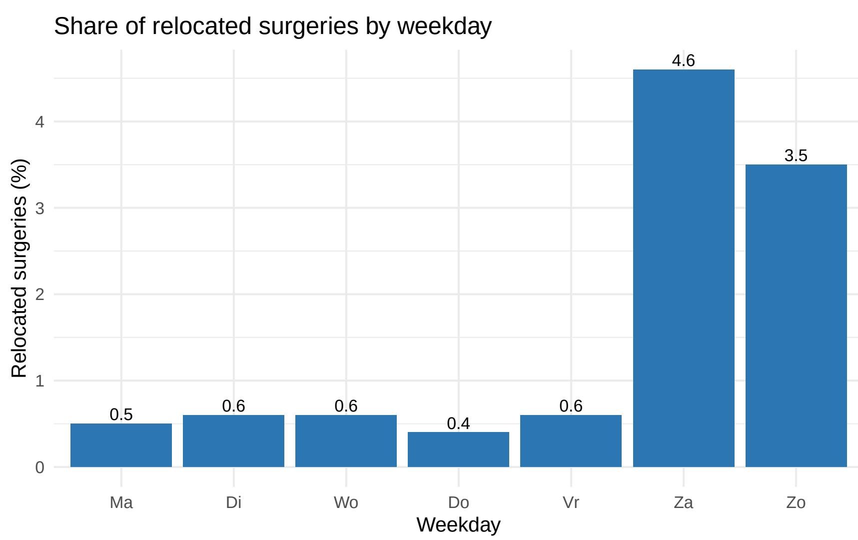

Figure 22: Share of relocated surgeries by weekday

The monthly percentage of surgeries that are relocated to a different operating room fluctuates between 0.25 and 1.2 percent, as shown in Figure 23. Despite short-term variation, no clear upward or downward trend is visible over the observed period. Occasional peaks above 1.0 percent indicate that some months experience more frequent reallocations, but these appear as isolated events rather than structural changes.

The relatively low and stable baseline confirms that room swaps are infrequent and mainly driven by temporary scheduling pressures or logistical constraints. The absence of a consistent trend over time suggests that the overall stability of room assignments at Genk has remained steady throughout the study period.

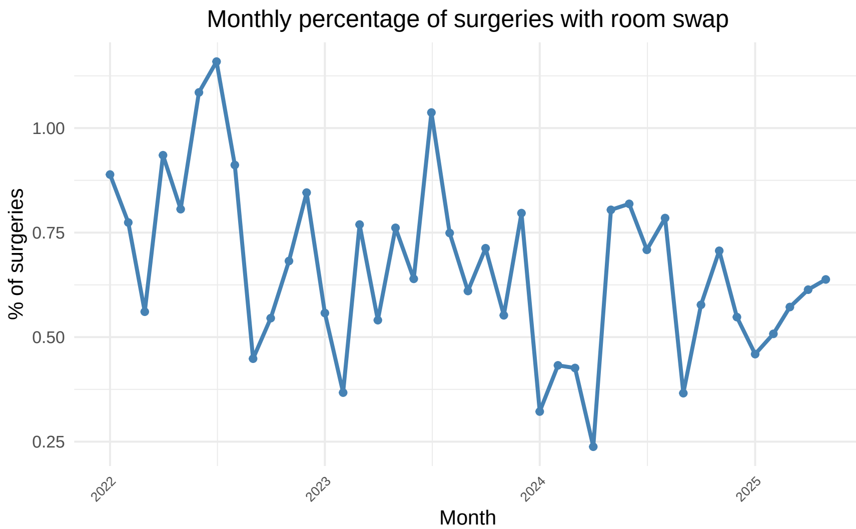

Figure 23: Monthly percentage of surgeries with room swap

Are specific rooms affected more than others?

The analysis shows that only a limited number of operating rooms are regularly involved in relocations, as summarized in Table 28. A few rooms act as central nodes that receive or exchange cases more frequently, while most others experience swaps only occasionally. The strongest interaction occurs between GO04 and GO05, which frequently trade procedures and appear to function as flexible backup locations. In addition, rooms such as GO09 and GO08 also show strong bidirectional exchanges, indicating similar functional overlap.

These consistent patterns indicate that room reallocations are not random but follow logical operational links between rooms with comparable setups or overlapping scheduling needs. Rather than signaling instability, such movements likely reflect the adaptive use of available capacity to maintain workflow continuity when minor scheduling conflicts occur.

Table 28: Top 10 planned ORs with the highest number of room swaps, showing the five most frequent new OR destinations (Genk).

PlannedOR	TotalMoved	Top1	Top2	Top3	Top4	Top5
GO05	55	GO04 (32, 58.2%)	GO11 (12, 21.8%)	GO06 (4, 7.3%)	GO03 (3, 5.5%)	GO01 (1, 1.8%)
GO04	47	GO05 (27, 57.4%)	GO11 (8, 17%)	GO03 (6, 12.8%)	GO08 (2, 4.3%)	GO02 (1, 2.1%)
GO09	26	GO08 (14, 53.8%)	GO11 (9, 34.6%)	GO05 (2, 7.7%)	GO13 (1, 3.8%)	NA
GO06	24	GO05 (11, 45.8%)	GO02 (4, 16.7%)	GO11 (3, 12.5%)	GO04 (2, 8.3%)	GO17 (2, 8.3%)

PlannedOR	TotalMoved	Top1	Top2	Top3	Top4	Top5
GO01	23	GO11 (10, 43.5%)	GO02 (5, 21.7%)	GO14 (2, 8.7%)	GO03 (1, 4.3%)	GO04 (1, 4.3%)
GO03	22	GO04 (8, 36.4%)	GO05 (5, 22.7%)	GO11 (5, 22.7%)	GO02 (2, 9.1%)	GO06 (1, 4.5%)
GO07	20	GO11 (5, 25%)	GO06 (3, 15%)	GO05 (2, 10%)	GO12 (2, 10%)	GO01 (1, 5%)
GO02	16	GO01 (7, 43.8%)	GO11 (4, 25%)	GO03 (2, 12.5%)	GO04 (1, 6.2%)	GO05 (1, 6.2%)
GO08	14	GO09 (10, 71.4%)	GO07 (2, 14.3%)	GO11 (2, 14.3%)	NA	NA
GO10	8	GO11 (5, 62.5%)	GO08 (2, 25%)	GO09 (1, 12.5%)	NA	NA

A small number of operating rooms receive the majority of relocated surgeries, as shown in Table 29. GO11 appears most frequently as destination, followed by GO05 and GO04, together accounting for the largest share of relocated cases. Several general-purpose rooms, including GO13 and GO08, also serve as recurring backup locations when schedule adjustments are required.

The distribution is highly concentrated, only a handful of rooms accommodate most relocations, while the remaining ORs receive very few transferred cases. This pattern indicates that specific rooms have more flexible capacity or broader compatibility with multiple procedure types. The low overall percentages confirm that relocations remain rare, but when they do occur, they are directed primarily toward a small subset of rooms designed or available to absorb unplanned activity.

Table 29: Top 15 actual operating rooms most frequently used as destinations after relocation (Genk).

ActualOR	ReceivedCases	SharePct
GO11	110	0.14
GO05	65	0.08
GO04	57	0.07
GO13	41	0.05
GO08	26	0.03
GO12	25	0.03
GO15	25	0.03
GO03	21	0.03
GO18	21	0.03
GO09	20	0.03
GO16	20	0.03
GO17	19	0.02
GO02	18	0.02
GO07	14	0.02
GO01	13	0.02

Procedures with the highest likelihood of being relocated mainly consist of short or outpatient interventions, as shown in Table 30. The top of the list is dominated by minor surgical activities such as plastic surgery cabinet procedures, small wound treatments, and standard outpatient operations. These procedures have relocation rates between roughly 3.6 and 9.4 percent, which is considerably higher than the overall campus average of 0.7 percent.

The pattern suggests that relocations are most frequent among short, low-complexity interventions that can be performed in multiple suitable rooms. This flexibility allows these cases to be reassigned when needed without disrupting the broader operating schedule. In contrast, major inpatient or specialized procedures appear far less often among relocated cases. Overall, the table shows that room swaps at Genk are concentrated in a small set of routine, easily transferable procedures.

Table 30: Top 15 procedure types with the highest share of relocated surgeries (Genk).

ProcedureName	n	SwapPct
KNIE WONDDEBRIDEMENT	32	9.38
WONDHECHTING	88	5.68
I & D VAN ADENOFLEGMONE	114	5.26
HEUP DRAINAGE BESMETTE WONDE	39	5.13
PERI-ANAAL ABCES	220	5
HAND WONDDEBRIDEMENT	61	4.92
POLS FRACTUUR REDUCTIE GESLOTEN	61	4.92
VOET WONDDEBRIDEMENT	61	4.92
TREPANATIE HEMATOOM	118	4.24
CURETTAGE	71	4.23
ELLEBOOG FRACTUUR REDUCTIE	49	4.08
VITRECTOMIE	51	3.92
VENTRIKEL DRAINAGE of EXTERNE DRAINAGE	155	3.87
ANDERE	52	3.85
BOORGATEN SUBDURAAL HEMATOOM	137	3.65

What is the operational impact of these reallocations?

Relocated surgeries differ from non-relocated ones in several aspects of timing and scheduling performance, as summarized in Table 31. For start times, relocated cases show a higher average delay among late starts (128 minutes versus 74.2 minutes) and a larger lead among early starts (68.9 minutes versus 34.3 minutes). This suggests that these surgeries are more often scheduled under atypical or time-constrained circumstances, which can result in greater variability in when they actually begin. The median values, however, remain closer together (55 versus 27 minutes for late starts, 28 versus 12 minutes for early starts), indicating that while some relocated surgeries start much later or earlier than expected, most are only moderately affected.

For procedure duration, both relocated and non-relocated surgeries show comparable deviations from their planned times. Average differences are around 28.9 versus 22.6 minutes for longer cases and 28.2 versus 22.1 minutes for shorter ones, suggesting that relocation itself does not substantially influence how long a surgery takes once it begins.

Overtime remains limited in both groups, with median values at zero. Mean overtime is slightly higher for relocated cases (8.2 versus 5.9 minutes), which indicates that relocations only marginally increase the likelihood of extended working hours.

Overall, the results indicate that relocations are associated with greater variability in start timing but not with systematically longer or shorter surgeries. Their operational impact therefore appears to be primarily related to scheduling and coordination rather than to the procedures themselves.

Table 31: Comparison of start and duration deviations (early/late, shorter/longer) and overtime between relocated and non-relocated surgeries (Genk).

Type	Metric	Mean	Median
Relocated	Late start	128	55
Not relocated	Late start	74.2	27
	—-		
Relocated	Early start	68.9	28
Not relocated	Early start	34.3	12
	—-		
Relocated	Longer duration	28.9	17
Not relocated	Longer duration	22.6	14
	—-		
Relocated	Shorter duration	28.2	15
Not relocated	Shorter duration	22.1	13
	—-		
Relocated	Overtime	8.2	0
Not relocated	Overtime	5.9	0
	—-		

Overtime is slightly more common among relocated surgeries than among those performed in their originally assigned room, as shown in Table 32. About 9.7 percent of non-relocated surgeries extend beyond 16:30, compared to 14.8 percent of relocated ones. This difference likely reflects that some relocated cases are scheduled later in the day or in response to last-minute adjustments, which can occasionally lead to minor overruns.

The percentages indicate that room reassignments are associated with a somewhat higher likelihood of overtime, although they do not represent a dominant driver of evening work at the campus level. Although relocations require coordination, their contribution to overall after-hours workload remains limited.

Table 32: Share of after-hours surgeries by relocation status (Genk).

SwapGroup	AfterHours	Percentage of surgeries with afterhours
Not relocated	7652	9.7
Relocated	77	14.8

The distribution of overtime duration for relocated surgeries, shown in Figure 24, is highly concentrated near zero, indicating that most of these cases finish within regular working hours or shortly afterward. A steep drop is visible beyond the first 30 minutes of overtime, and only a few relocated surgeries extend beyond two hours. The long but sparse tail confirms that extreme cases are rare.

This pattern supports the earlier finding that relocations mostly involve short procedures that can be completed quickly once a new room becomes available. The limited overtime associated with these cases suggests that room reassignments are generally managed efficiently and do not significantly contribute to prolonged evening work. In operational terms, relocated surgeries appear to represent flexible workload adjustments rather than sources of additional delay.

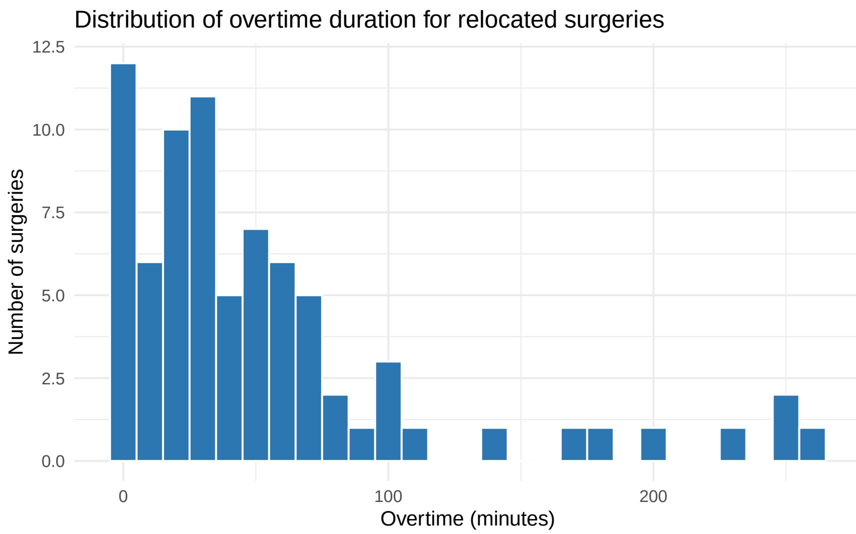

Figure 24: Distribution of overtime duration for relocated surgeries

How do urgent cases disrupt elective planning?

When and where do urgent surgeries occur?

The majority of surgeries at Genk are elective, representing about 85.4 percent of all recorded cases, as shown in Table 33. Non-elective or urgent surgeries make up the remaining 14.6 percent of the total. This indicates that while most activity follows a planned schedule, a meaningful portion of cases still arises from unplanned or time-sensitive demands.

The relative size of the non-elective segment highlights its operational significance. Even though urgent cases form a minority, their unpredictable timing can affect the stability of daily planning and room allocation. Understanding when and where these non-elective procedures occur provides an important basis for evaluating how they interact with elective activity and whether they contribute to schedule deviations or after-hours work.

Table 33: Distribution of surgeries by urgency classification (Genk).

UrgencyType	Total	SharePct
electief	67736	85.4
niet-electief	11616	14.6

The start times of non-elective surgeries, shown in Figure 25, display a clear concentration during daytime hours, with a pronounced peak between 13:00 and 15:00. This pattern indicates that the majority of urgent procedures are performed within or near regular operating hours, often overlapping with scheduled elective activity. A smaller but steady level of activity extends into the evening, reflecting cases that arise later in the day and require prompt intervention.

Night-time activity is relatively limited, with few surgeries starting between midnight and early morning. Overall, the distribution suggests that non-elective surgeries are not evenly spread across the 24-hour period but cluster strongly in the afternoon and early evening, when operating room capacity is already in high demand.

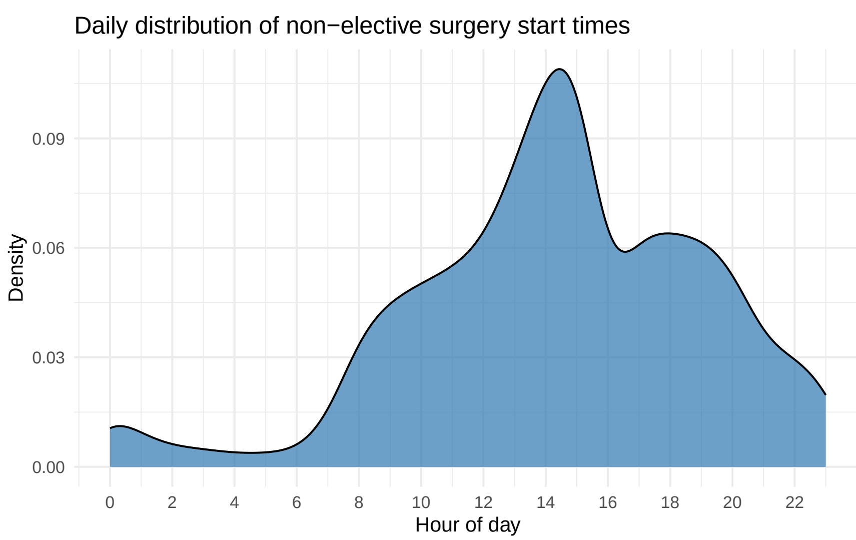

Figure 25: Daily distribution of non-elective surgery start times

Across all weekdays, non-elective surgeries follow a consistent hourly pattern, as illustrated in Figure 26. Most activity occurs between late morning and early evening, with peaks around 13:00 to 15:00 on nearly every day. This confirms that urgent procedures are frequently performed during regular daytime hours. The overlap with elective schedules suggests that urgent cases often need to be accommodated within existing capacity rather than outside normal shifts.

During weekends, the temporal pattern flattens slightly, but a similar afternoon concentration remains visible. Saturday and Sunday both show sustained activity throughout the day and into the evening, reflecting the continuous need for urgent surgical care even when elective operations are limited. Overall, the figure highlights that while urgent surgeries occur daily, their timing is strongly clustered in the same hours as routine operating room use, which can create additional scheduling pressure during peak times.

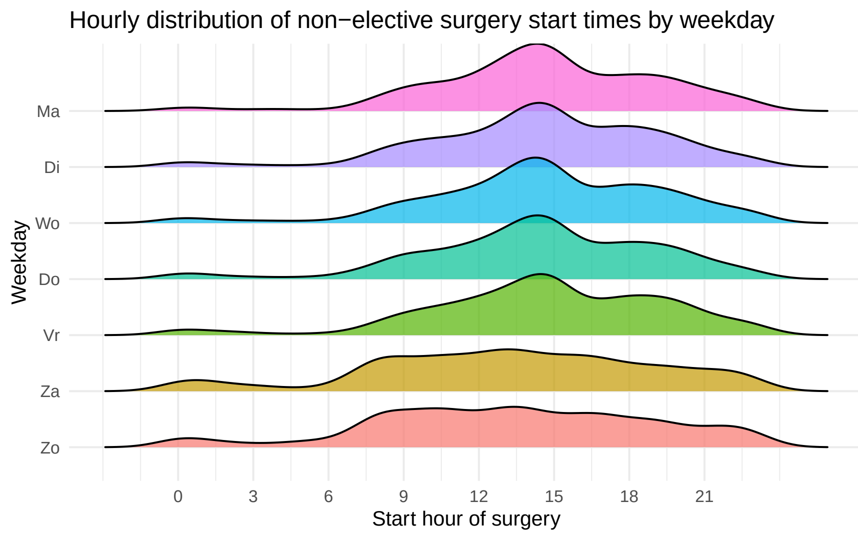

Figure 26: Hourly distribution of non-elective surgery start times by weekday

Elective surgeries dominate the activity on weekdays, as shown in Table 34, accounting for roughly 86 to 88 percent of all cases from Monday to Friday. The share of non-elective procedures remains relatively stable during the workweek, between 11.6 and 13.8 percent, indicating a consistent and predictable pattern of urgent demand. This stability suggests that urgent cases are routinely managed alongside scheduled operations without major fluctuations across days.

A different picture emerges during the weekend. On Saturday, nearly 65 percent of all surgeries are non-elective, reflecting a clear shift toward urgent activity when elective scheduling is minimal. On Sunday, this proportion remains high at 60.3 percent. The data confirm that weekday operations are primarily planned and elective in nature, while weekend activity is driven mainly by urgent and unplanned cases. This distinction highlights the importance of maintaining dedicated flexibility in scheduling and staffing to handle urgent demand without disrupting regular weekday operations.

Table 34: Weekday distribution of elective vs. non-elective surgeries (Genk).

Weekday	Elective (n)	Non-elective (n)	Elective (%)	Non-elective (%)
Ma	12274	1958	86.2	13.8
Di	12810	1940	86.8	13.2
Wo	13970	1844	88.3	11.7
Do	14003	1841	88.4	11.6
Vr	13527	2088	86.6	13.4

Weekday	Elective (n)	Non-elective (n)	Elective (%)	Non-elective (%)
Za	579	1075	35	65
Zo	573	870	39.7	60.3

Do urgent cases drive overtime?

Urgent cases contribute to overtime work but are not its primary driver, as shown in Table 35. While the share of non-elective surgeries extending beyond the shift is higher (18.2 percent) than for elective ones (8.3 percent), the overall difference remains moderate. This indicates that both types of activity regularly continue after the shift, although elective cases represent the majority in absolute numbers due to their much larger total volume.

Average overtime per case is somewhat higher for non-elective surgeries, at 10.7 minutes compared with 5 minutes for elective ones. The 95th percentile follows a similar pattern, reaching 69 minutes for urgent cases versus 29 minutes for elective ones. This suggests that urgent procedures, while less frequent, are more likely to require longer extensions when they do occur.

Overall, the results imply that unplanned demand does contribute to extended operating hours, but most evening work still stems from elective cases that overrun their planned schedules rather than from late-arriving emergencies.

Table 35: After-hours frequency and overtime duration by urgency type (Genk).

Urgency type	Total (n)	After-hours (n)	After-hours (%)	Mean overtime (min)	95th percentile (min)
Elective	67736	5620	8.3	5	29
NonElective	11616	2109	18.2	10.7	69

Do urgent cases interrupt elective planning?

The results in Table 36 show that the frequency of room swaps among elective surgeries is low in both groups but somewhat higher when overlap occurs. For surgeries not affected by an overlap, 0.3 percent involved a room swap, compared to 0.6 percent among overlapped cases.

This pattern indicates that while urgent procedures occasionally coincide with scheduled elective activity, they may slightly increase the likelihood of room reallocations, although the overall impact remains limited. Instead of systematically triggering swaps, most overlaps appear to be managed through local coordination or minor timing adjustments rather than by physically moving elective cases to different operating rooms. Overall, urgent interventions create some scheduling pressure but do not substantially disrupt room allocation patterns within the elective program.

Table 36: Room swap frequency among elective cases: overlapped vs. not overlapped.

Overlapped	N	RoomSwaps	ShareSwapped
No	64405	217	0.3
Yes	3331	20	0.6

As shown in Table 37, overlaps between urgent and elective surgeries occurred on 858 out of 1247 observed days, corresponding to 68.8 percent of all days. This high frequency indicates that schedule interference is not an exceptional event but a regular characteristic of daily activity. While individual overlaps may vary in magnitude and operational impact, their near-daily occurrence highlights the continuous need for flexibility in resource allocation and real-time coordination between planned and urgent care.

Table 37: Days with at least one urgent–elective overlap in the same OR.

OverlapDays	TotalDays	SharePct
858	1247	68.8

The proportion of elective surgeries affected by urgent overlaps fluctuates modestly over time, as shown in Figure 27. Monthly values generally range between 3.7 and 6.5 percent. A gradual upward trend is visible throughout 2023 and early 2024, followed by a temporary decline in the second half of 2024. Despite these month-to-month variations, the overall level remains relatively stable, suggesting that the frequency of urgent cases interfering with scheduled activity is a persistent feature of daily operations rather than a short-term anomaly.

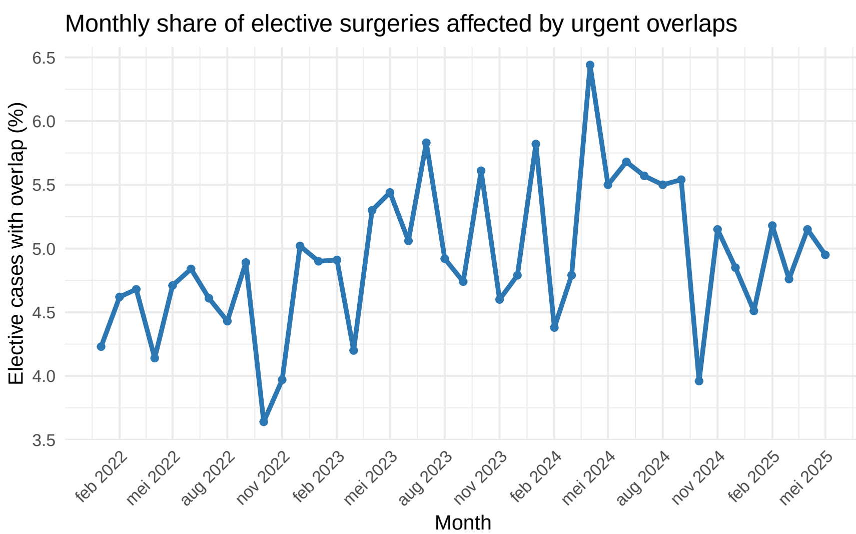

Figure 27: Monthly share of elective surgeries affected by urgent overlaps

As shown in Figure 28, overlaps between urgent and elective surgeries occur predominantly during daytime hours. The frequency increases sharply after 08:00, peaks between 13:00 and 15:00, and then declines toward the evening. Only a small number of overlaps are observed during the night and early morning. This temporal pattern mirrors the general distribution of non-elective surgery starts and confirms that most schedule conflicts arise when operating room utilization is already at its highest. The clustering of overlaps in the middle of the day suggests that urgent admissions frequently coincide with the busiest elective periods, intensifying competition for operating room capacity during standard working hours.

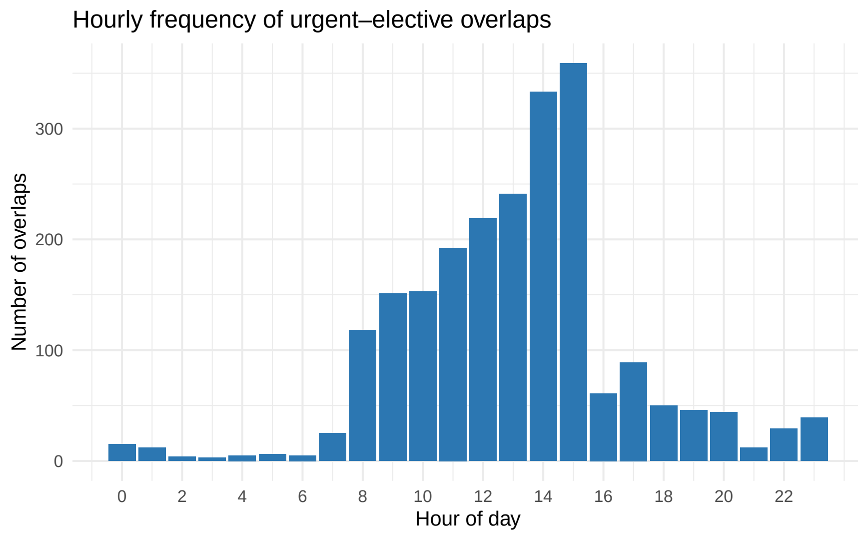

Figure 28: Hourly frequency of urgent–elective overlaps

Figure 29 shows that elective surgeries affected by an urgent overlap consistently start later than those without overlap. The gap is largest in early 2022, when mean start delays for overlapped cases frequently exceeded 70 minutes, compared with around 30 minutes for non-overlapped cases. Although the difference narrows over time, the delay for overlapped surgeries remains visibly higher throughout the entire period.

The recurring peaks indicate that overlaps are linked to periodic scheduling pressures rather than isolated disruptions. Overall, the figure confirms that the occurrence of urgent cases has a measurable and sustained effect on elective punctuality, particularly during months of high operating room utilization.

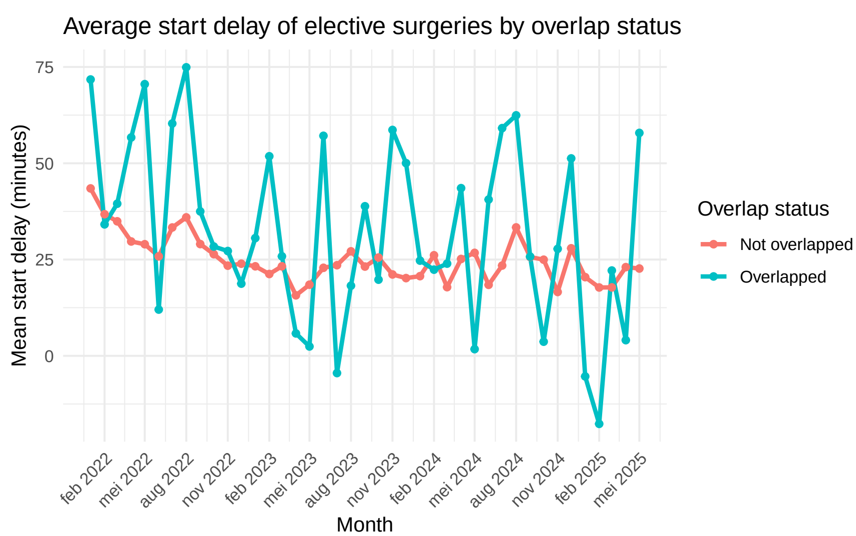

Figure 29: Average start delay of elective surgeries by overlap status

As shown in Table 38, the frequency and impact of overlaps vary markedly across operating rooms. The emergency-designated room GO11 shows the highest number of overlap events, with 475 elective cases affected, representing 15.2 percent of all elective activity in that room. This confirms its frequent dual use for both scheduled and urgent procedures. Several other rooms, including GO06, GO05, and GO04, also record substantial numbers of overlaps, though at lower relative shares.

Median start delays reveal a heterogeneous pattern. In most rooms, elective cases with overlaps tend to start later than those without, indicating disruption to planned schedules. However, a few rooms such as GO17, GO13, and GO12 show negative median differences, suggesting that rescheduling or case reassignment may occasionally allow elective surgeries to start earlier than initially planned.

Overall, the data illustrate that the operational impact of urgent cases is concentrated in a limited number of high-volume rooms, where the coexistence of elective and non-elective activity is most frequent.

Table 38: Operating rooms with urgent–elective overlap.

PlannedOR	Count	n_elective	Percentage	Med StartDelay (No Overlap)	Med StartDelay (Overlap)
GO11	475	3120	15.2	28	60
GO06	324	4760	6.8	5	3
GO05	308	3585	8.6	8	13
GO04	268	3561	7.5	5	6
GO17	232	5210	4.5	8	-6.5
GO18	199	5031	4	8	0
GO07	198	3916	5.1	10.5	15.5
GO03	159	3676	4.3	7	6
GO09	148	2214	6.7	5	7
GO02	143	3272	4.4	8	13
GO13	131	2813	4.7	1	-7
GO12	122	2607	4.7	2	-4
GO14	116	6701	1.7	15	20.5
GO01	115	6307	1.8	14	9
GO15	114	4055	2.8	8	4
GO16	108	3937	2.7	8	-3
GO08	95	1499	6.3	12	14
GO10	76	1472	5.2	-2	-1

How stable is the surgical schedule in execution?

Beyond accurate planning, the stability of the realised schedule is a key indicator of operational performance in the operating theatre. Even when planned start times, durations, and room assignments are correct, day-to-day execution may deviate because of unforeseen delays, emergency insertions, or coordination issues.

This section evaluates how consistently the planned schedule is maintained throughout the day by focusing on two complementary aspects:

- *Shift changes*, where surgeries are executed in a different shift than originally planned, indicating how often the daily programme must be adjusted in real time.
- *Idle or gap times*, representing the pauses between consecutive surgeries within the same room, which reflect how efficiently the available capacity is utilised.

Together, these indicators capture the operational resilience of the OR complex and the ability to absorb disruptions and maintain smooth throughput despite variable circumstances. High rates of shift changes or long idle periods point to latent inefficiencies in daily planning and coordination, whereas stability across these metrics signals a well-synchronised workflow.

How often are surgeries moved to another shift?

Changes between planned and actual shifts occur when surgeries start significantly later than scheduled or are postponed to a later block of the day. The actual shift is determined based on the observed start time of each surgery relative to the official shift boundaries used at the operating department (day: 08:00–16:30, evening: 16:30–22:00, night: 22:00–08:00).

At Campus Genk, 4151 surgeries, or 5.2 percent of all recorded cases, were performed in a different shift than originally planned, as shown in Table 39. For these surgeries, the average start delay was 398.3 minutes, indicating that such deviations typically involve a substantial postponement within the daily schedule. Interestingly, the average deviation in procedure duration was –22.3 minutes, meaning that surgeries executed in another shift tend to be somewhat shorter than their planned estimates once they begin.

The mean overtime for these cases was 10.4 minutes, showing that most shift changes result from late starts or, in some cases, procedures that exceed their planned duration, rather than extensive spillovers deep into the next shift. Although these cases represent only a small fraction of total surgical activity, their magnitude and timing can still disrupt staffing and sequencing for subsequent sessions.

Table 39: Cases performed in a different shift than planned (Genk).

n	MismatchCount	Share moved (%)	Mean Start Delay (min)	Mean Duration Diff (min)	Mean Overtime (min)
79352	4151	5.2	398.3	-22.3	10.4

As shown in Figure 30, the monthly trend shows visible variability but no structural deterioration over time. In early 2022, the share of surgeries performed in a different shift was relatively high, fluctuating around 6 to 8.5 percent. Over the following months, the proportion gradually declined to around 3 to 5 percent, where it remained throughout 2023 and 2024. Occasional short-lived peaks appear, but these fluctuations are modest and do not indicate a systematic increase.

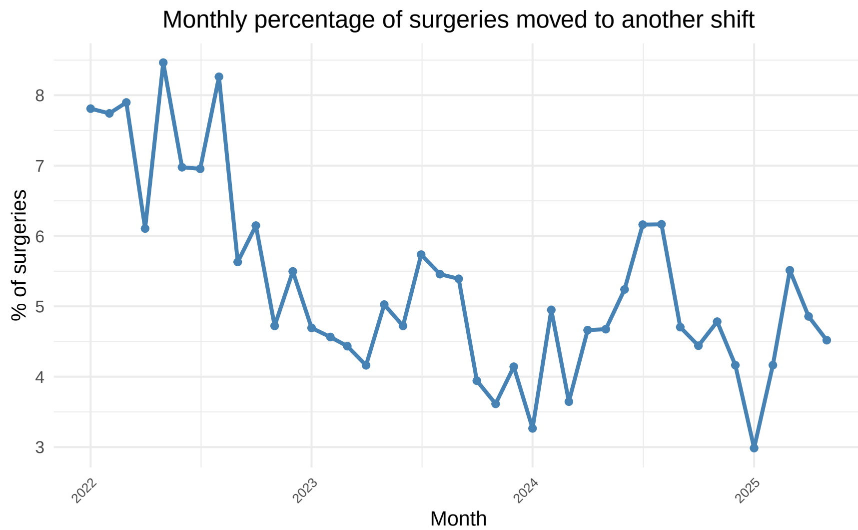

Figure 30: Monthly percentage of surgeries moved to another shift

Overall, the data suggest a moderate improvement in schedule adherence since 2022, followed by a period of relative stability. The month-to-month variation likely reflects normal fluctuations in workload and case mix rather than structural scheduling issues.

How much idle time occurs between consecutive surgeries?

Idle or gap time represents the short pause between the end of one surgery and the start of the next within the same operating room during the regular day shift (08:00–16:30). It captures how efficiently rooms are prepared for the next case and how well surgical schedules are synchronised in daily practice. For each room-day combination, the gap time measures either the delay between shift start and the first case or the interval between consecutive cases, excluding breaks longer than 60 minutes that reflect planned downtime.

At Campus Genk, the average idle time between consecutive surgeries is 9.9 minutes, while the median is 8 minutes, as shown in Table 40. The relatively small difference between both measures indicates that turnover between cases is generally quick and consistent. The interquartile range of 7 minutes further confirms that most transitions take place within a narrow time window. The 95th percentile of 25 minutes shows that only a small portion of cases experience substantially longer changeovers, while the maximum observed idle time of 60 minutes marks the applied cutoff for excluding planned downtime.

Table 40: Summary statistics of idle time between consecutive surgeries (minutes).

Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9
9.9	8	7.9	7	0	60	25	42	58

As shown in Figure 31, the histogram confirms that most idle intervals are very short. Nearly half of all cases show a pause of less than 5 minutes, and about three quarters fall below 10 minutes. The frequency then decreases rapidly, with only a small share of transitions exceeding 20 to 30 minutes. The right-hand tail remains thin, as gaps longer than one hour are excluded from the analysis.

Overall, the distribution demonstrates that turnover between cases is highly efficient and consistent across rooms. Short idle periods dominate the workflow, while occasional longer gaps occur infrequently and remain within operational limits rather than indicating systematic underutilisation.

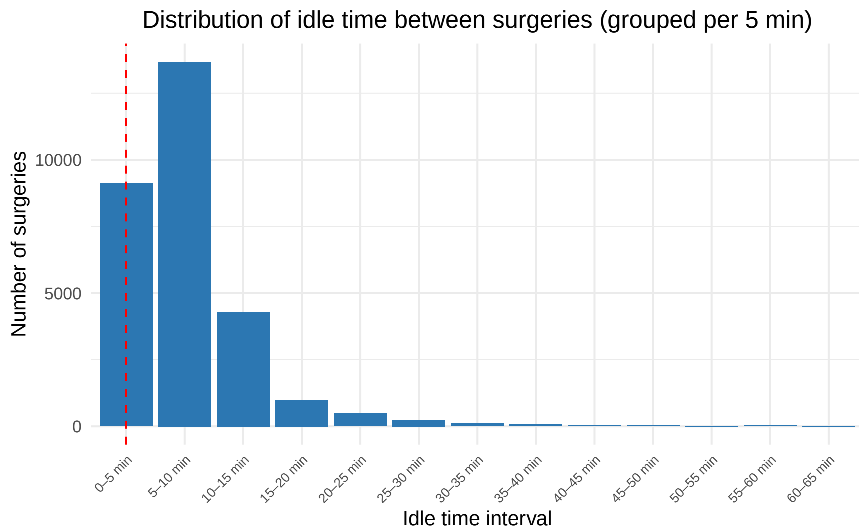

Figure 31: Distribution of idle time between surgeries (grouped per 5 min)

How does idle time vary across operating rooms?

As shown in Figure 32, the mean and median idle times differ moderately across operating rooms at Campus Genk. In all rooms, the median idle time is shorter than the mean, confirming that most intervals between consecutive surgeries are brief while a limited number of longer pauses raise the average. The consistent gap between both measures illustrates the right-skewed nature of idle-time distributions, where short intervals dominate but longer ones remain present.

The absolute differences between rooms are modest. Median values generally stay within a narrow range, indicating comparable turnover performance across the OR complex. The highest idle times are observed in rooms such as GO11, where both mean and median values are clearly above average. GO10 shows a higher mean but a more moderate median, suggesting that a limited number of longer gaps drive its average rather than consistently slower turnover. In most other rooms, mean and median idle times cluster at lower levels, reflecting uniform and efficient transitions between cases.

Overall, idle time across rooms is contained and stable. Occasional longer gaps appear to result from isolated extended intervals rather than systematic inefficiencies. The figure therefore highlights a consistently well-coordinated workflow, with only minor room-to-room variation in turnover performance.

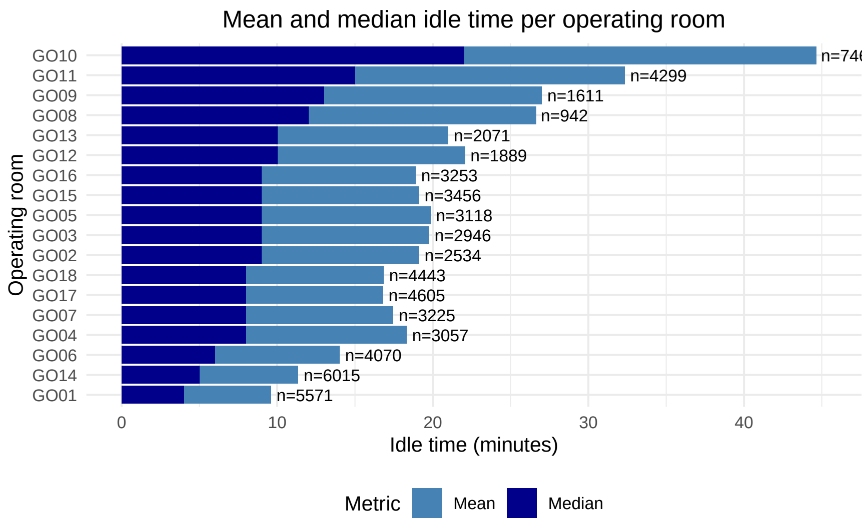

Figure 32: Mean and median idle time per operating room

As shown in Table 41, the standard deviation (SD) and coefficient of variation (CV) quantify how much idle times fluctuate around their typical value for each operating room, sorted by median idle time. Across most operating rooms, the SD ranges between about 5 and 11 minutes, while the CV values generally fall between 60 and 80 percent, indicating moderate relative variability. Turnover intervals are typically short but not perfectly consistent.

Rooms such as GO11 and GO08 stand out with higher SD and CV values, reflecting greater spread in idle times and therefore less uniform turnover performance. GO10 shows a relatively high SD but a lower CV, indicating that while absolute variation is larger, it remains more limited relative to its mean idle time. In contrast, rooms like GO03, GO02, GO15, and GO16 combine lower SD and CV values, suggesting that their turnover durations are more concentrated around the median.

Toward the bottom of the table, a few rooms, including GO14 and GO01, display higher CV values despite relatively low SDs. This pattern results from the ratio-based nature of the CV, which becomes large when the average idle time is small.

Overall, variability in idle time differs meaningfully between rooms but remains within a limited range for most of the OR complex. The results confirm that short turnover periods dominate, while a few rooms exhibit broader relative dispersion, reflecting either shorter baseline idle times or more variable scheduling conditions.

Table 41: Standard deviation and coefficient of variation of idle time per operating room (minutes), sorted by median idle time.

ActualOR	Cases	Standard deviation	Coefficient of variation
GO10	746	10.5	46.2
GO11	4299	10.8	62.5
GO08	942	10.4	70.8
GO09	1611	8.2	58.3
GO12	1889	8.7	72.2
GO13	2071	7.2	65.2
GO05	3118	8.3	76.8
GO03	2946	7	65.3
GO04	3057	8.1	78.8

ActualOR	Cases	Standard deviation	Coefficient of variation
GO02	2534	6.6	64.8
GO15	3456	6.7	66.1
GO16	3253	6.2	62.5
GO07	3225	7.1	74.8
GO18	4443	6.5	73.7
GO17	4605	6.5	74.1
GO06	4070	6.3	78.6
GO14	6015	5.2	81.9
GO01	5571	5	89.4

Conclusion

This in-depth analysis of Campus Genk provides a detailed and data-driven view of how surgical activity evolves from planning to execution within the operating theatre. The report systematically examines timing accuracy, room allocation, after-hours activity, urgent case dynamics, and turnover performance, offering insight into both the strengths and operational challenges of daily surgical practice.

Across the different sections, the results show that planned and observed durations are generally well aligned, though variability between procedures remains considerable. Start-time delays are common and tend to accumulate throughout the day, which can lead to later-than-expected finishes. While the overall schedule performs reliably, these small deviations underscore the operational sensitivity of tightly sequenced programs.

After-hours activity represents a recurring but contained part of the workload, with about 9 to 10 percent of surgeries extending beyond regular hours. Most overruns are limited in duration, yet a small number of complex procedures still account for a large share of total overtime. Room assignments remain highly stable, as only about 0.7 percent of surgeries are relocated, and these typically concern short, low-complexity procedures that have minimal workflow impact.

Urgent cases make up roughly 14 percent of all surgeries and frequently overlap with elective activity. Although their overall share is modest, such overlaps occur on nearly 70 percent of days, emphasizing their continuous operational relevance. While urgent cases can occasionally delay elective starts, they rarely lead to extensive rescheduling or room reallocations, and they are not the primary driver of overtime, indicating that daily coordination between planned and unplanned care is well managed.

The analysis of shift transitions and idle times further illustrates an efficient process. Around 5 percent of surgeries are performed in a different shift than initially planned, a small but operationally relevant proportion. Idle periods between surgeries are short and consistent across rooms, with a median of 8 minutes and moderate variability. Turnover is therefore fast and well controlled, though a few rooms display broader distributions that point to more diverse scheduling conditions.

Altogether, this document offers a comprehensive quantitative picture of operating room performance at Campus Genk. It shows how surgical planning translates into real-world execution, balancing predictability and flexibility in a complex, high-throughput environment. The findings confirm an overall efficient system, with targeted opportunities to further stabilise start times, reduce minor delays, and maintain smooth coordination between elective and urgent surgical activity.